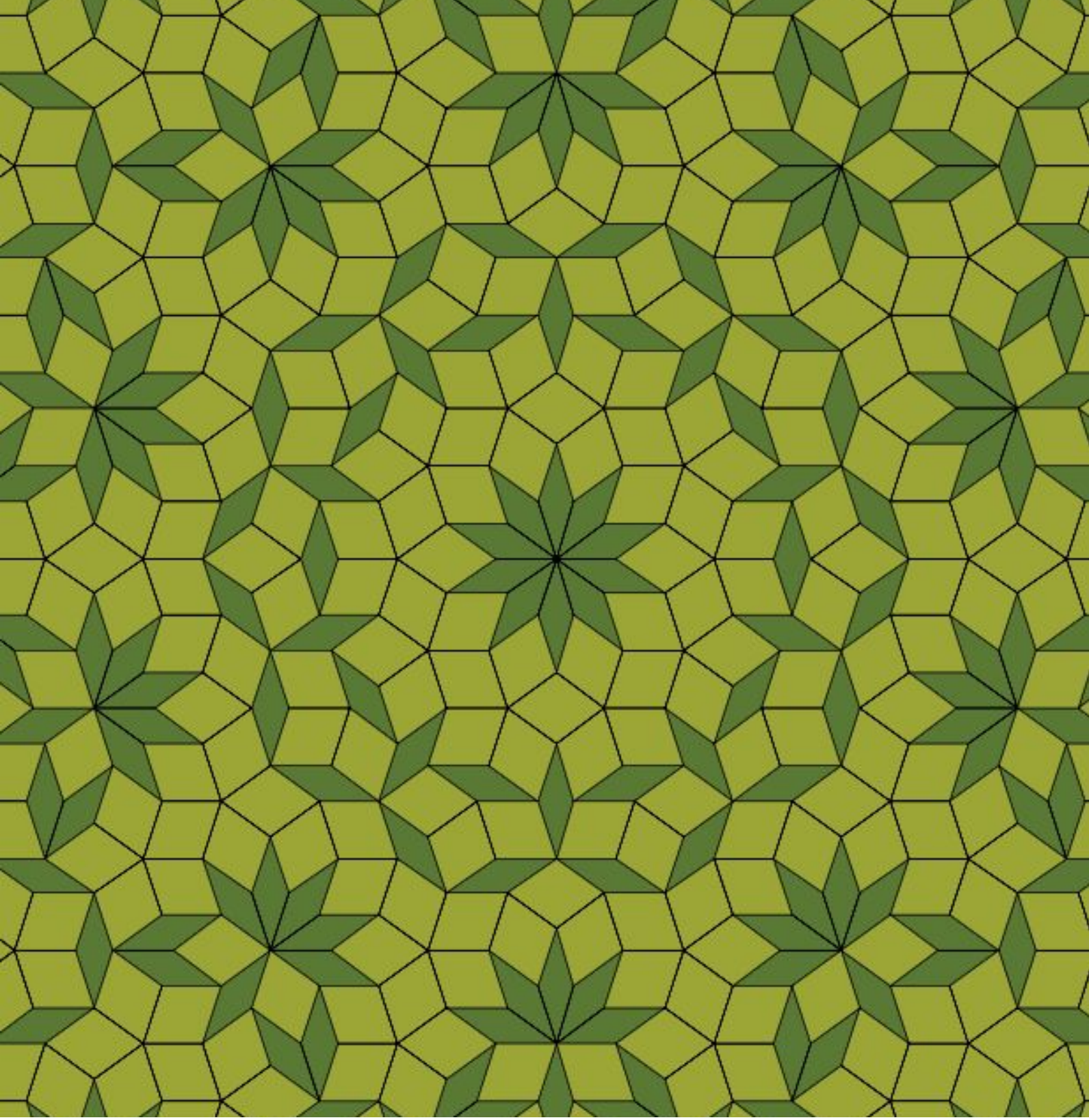


THE PENROSE MAGAZINE

STEM

Issue 11: Competition Issue, 12 Nov 2026

Welcome to the Climate in STEM Competition Issue of Penrose Magazine!

Penrose is a STEM magazine where we hope to establish a community of young people who are passionate about STEM and want to further their knowledge beyond the curriculum. This installment features the winning articles from the Penrose Magazine Summer 26 Competition revolving around the theme 'Climate in STEM.'

Note: Each article is displayed as it was submitted. The only changes that may have been made by the Penrose Magazine team were to citations to meet our formatting standards.

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The Cryosphere and Climate Change: When Arctic Ice Melts, the Planet Warms Faster

Arctic sea ice acts as a mirror by reflecting insolation into space and reducing oceanic solar absorption. When this reflective ice cover melts, it initiates the ice-albedo feedback loop [1]. Albedo is the fraction of insolation, or incoming solar radiation, that a surface reflects into space [2]. High-albedo surfaces (e.g., ice, clouds) reflect a large amount of sunlight into the atmosphere. Darker surfaces (e.g., forests, oceans) exhibit a low albedo and absorb more insolation, raising surface temperatures [2]. Earth's planetary albedo reflects 30% of insolation into space, while the remaining 70% is absorbed by the atmosphere, landmasses, and oceans [3]. Arctic sea ice reflects most insolation, but as temperatures rise, the ice melts to expose low-albedo ocean water, which absorbs 4.5 to 9 times as much as the ice it replaced, further intensifying the process [4]. Sea-ice albedo feedback contributes to Arctic amplification, causing the Arctic to heat up 3.8 times faster than the global average [5].

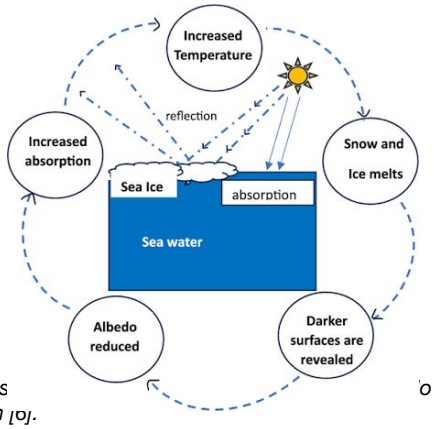


Fig. 1. This o feedback mechanism [6].

The link between ice loss in the Arctic and greenhouse gas emissions is linear, with satellite records showing September Arctic sea ice declining in proportion to carbon dioxide emissions, losing roughly 3.0±0.3 square meters of ice for every metric ton of cumulative anthropogenic CO_2 emissions since 1953 [7].

Arctic Sea-Ice Cover

Thick multiyear ice, which has survived at least one summer melt season, now covers less of the Arctic Ocean compared to the 1980s. The National Oceanic and Atmospheric Administration reported that 95,000 km² of ice older than four years remained at the September 2025 sea-ice minimum, which is a 95% decline from the 1985–2004 average [8]. Since multiyear ice is more resistant to summer melt, losing older ice leaves a greater share of the Arctic covered by vulnerable first-year ice.

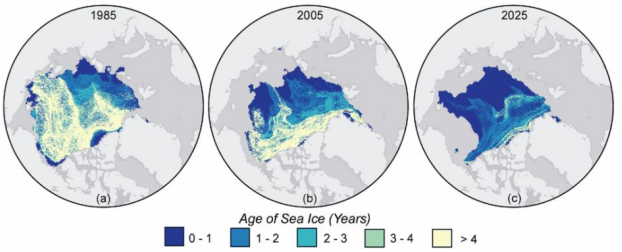


Fig. 2. Sea ice age during the week of the annual minimum ice extent in (a) 1985, (b) 2005, and (c) 2025 [8].

The Arctic's ice cover demonstrates the growing importance of ice-albedo feedback. In March 2026, Arctic sea ice likely reached an annual maximum of 14.29 million km², statistically tying with 2025 for the lowest maximum in the 48-year satellite record. A single record-low year does not prove a climatic pattern, but this event is consistent with the long-term decline observed since satellite monitoring began in 1979 [9].

The same warming process is accelerating ice loss in the Greenland Ice Sheet, which is losing about 234 billion tons of ice per year [10]. Furthermore, as shown in Fig. 3, the complementary NASA satellite missions GRACE-FO (blue) and ICESat-2 (green) track the changing mass and volume of the Greenland Ice Sheet, respectively. In 2025,

the Greenland Ice Sheet lost an estimated 129±50 gigatons of mass. Although this was less than the 2003–2024 average annual loss of 219±16 gigatons, it continued a pattern of net annual ice loss that has persisted since the 1990s [11].

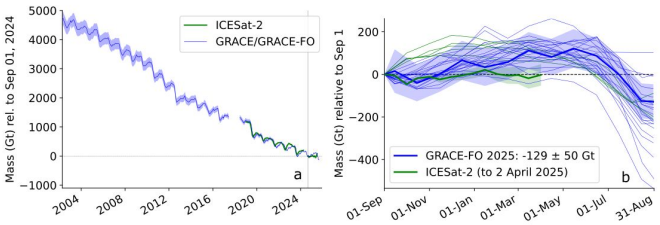


Fig. 3. Shows satellite data tracking the Greenland Ice Sheet's changing mass. Graph (a) displays the complete data history from the GRACE-FO and ICESat-2 satellite missions, including shaded areas to represent measurement uncertainty. Graph (b) compares the 2025 mass balance year (bold lines) against the 2003-2024 historical average (thin lines) [11].

Melt Ponds

As summer melting becomes widespread, meltwater collects in depressions on the ice surface, forming melt ponds. They form each spring and refreeze in autumn, and are a direct result of ice-albedo feedback [12]. At the height of summer, ponds can cover up to half the Arctic sea area [13].

In the Greenland Ice Sheet, the same process behaves differently. Melt ponds at the surface of the ice sheet can vanish in a few hours because the water flows off from the melt pond through vertical drainage shafts known as moulins [10]. NASA measured the Greenland Ice Sheet moulins down to 1,500 feet, and it was verified that the number of these moulins has been multiplying due to accelerated surface melting, although moulins are constantly merging each melt season [14].

Meltwater sliding down deep holes in the ice sheet reduces friction at the bedrock and displaces sediment, speeding up ice flow towards the ocean. For example, an outlet glacier on the west coast of Greenland called the Jakobshavn Glacier has been draining roughly 8% of the ice sheet. The glacier has recently doubled in flow speed towards the Ilulissat Icefjord during the last decade, and the flow speed of other glaciers in the area has increased by 50% [14]. Furthermore, global sea levels have risen by 9 inches since 1880, with the rise mostly due to a combination of meltwater from glaciers and ice sheets and thermal expansion of seawater as it warms [15].

To summarize, meltwater contributes to rising sea levels in 2 ways. The first is directly contributing to sea level rise when water from the ice finds its way to the ocean, and the second is the water's ability to speed up the flow of glacial ice towards the ocean [10].

Seasonal Ocean Heat Storage

Arctic sea ice is becoming a more seasonal phenomenon due to global warming. Across the Arctic, refreezing occurs about a month later than 40 years ago [16]. For example, in the Canada Basin, the delay is associated with a rise in seasonal heat stored in surface waters: cooling to the freezing point requires more heat loss to the atmosphere in autumn [17]. Seasonal heat storage is the thermodynamic process by which the Arctic Ocean absorbs massive amounts of solar energy during the summer, stores it in a mixed surface layer, which is typically down to 50-200 meters in the upper ocean, and then slowly redistributes that same heat after solar radiation declines [18].

In the western Arctic, this stored heat resides in a near-surface temperature maximum at about 20-30 m deep. It is formed when warm surface water is covered by colder meltwater. Due to its shallow depth, mixing can release the stored heat back into the ice-ocean layer and delay refreezing [19].

Solar radiation hits open water exposed by retreating ice, lowering surface albedo and increasing heat input into the upper mixed layer of the ocean. Instead of immediately warming the air, a large part of this absorbed solar energy is stored in the upper ocean's thermal capacity during the months of peak insolation. As the polar atmosphere cools in the autumn and winter, the temperature gradient inverts, and the ocean

discharges the heat it stored during the summer into the atmosphere [20]. The exact timing of that heat release matters. For example, in a September 2020 Beaufort Sea comparison of satellite-derived sea surface temperatures with direct ocean observations to examine the delay in autumn freeze-up, open water retained an additional 4.2×10^7 J m^{-2} of thermal energy relative to adjacent ice-covered water just 6 km away. This was enough to delay refreezing by about 4.4 days of continuous heat loss [17].

This regional heat reservoir, which is when a macro-environmental body (like an ocean) stores massive amounts of thermal energy, is different from the global ocean heat uptake, which is driven by Earth's overall energy imbalance. While the global ocean absorbs most of the excess heat added to the climate system, the primary issue in the Arctic is the timing of regional uptake and release [17].

Solutions

Melting sea ice is affecting the world by raising sea levels, reinforcing climate change, and putting Arctic animal species at risk [21]. For example, the loss of Arctic sea ice as a habitat is forcing polar bears onto land, where they fast for longer periods and produce smaller cub litters. It also drives walruses ashore in large numbers, where overcrowding leads to deadly stampedes [22]. To prevent Arctic ice from disappearing completely, emissions must limit global warming from exceeding 1.5°C (2.7°F) [23]. Geoengineering approaches include pumping seawater onto ice or draining melt ponds so water can refreeze. In Real Ice's Cambridge Bay, Nunavut trial, researchers pumped seawater through nearly 200 holes during the 2023-2024 winter and treated approximately 250,000 m^2 of sea ice, with a mean thickness of 1.9 m, increasing in thickness by about 0.46 m after the experiment [24]. Similarly, hollow silica-glass microspheres have been proposed for deployment over small Arctic areas to reduce absorbed insolation, but the approach still requires environmental assessment [25].

Conclusion

Arctic ice loss is a consequence and amplifier of global warming. As reflective ice breaks down into dark ocean water, more solar energy is absorbed and stored in the upper ocean. That heat gets released in ways that delay refreezing and promote further melt. Melt ponds and the loss of multiyear ice strengthen this process. Reducing greenhouse gas emissions is the only permanent solution for limiting sea ice albedo feedback and preserving Arctic ice.

By Chandrala Kulkarni

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Using Machine Learning to Predict Climate-Driven Disease Outbreaks

As global warming expands the ecological habitats of disease-carrying vectors, mosquito-borne and tick-borne illnesses are surfacing in historically unaffected regions [1]. Traditional epidemiological surveillance often lacks the predictive agility needed to map these rapid transitions. However, the convergence of data, science and medicine introduces a proactive paradigm through machine learning. By integrating heterogeneous datasets (e.g. satellite land-surface temperatures, relative humidity patterns and local vector lifecycles), neural networks can simulate and forecast outbreak hotspots weeks in advance. This paper goes into how predictive algorithms can act as vital public health instruments, bridging environmental data science with preventative medicine to reduce the burden on healthcare systems caused by a warming planet.

Introduction

The direct impact of climate change extends far beyond rising sea levels and extreme weather anomalies; it is fundamentally altering the global landscape of infectious disease transmission. Vectors such as *Aedes Aegypti* mosquitoes and various tick species are highly sensitive to microscopic environmental shifts.

Incremental increases in average global temperatures, as well as unpredictable precipitation cycles, dramatically accelerate vector incubation rates and expand their viable geographic boundaries into high-latitude zones [1]. Consequently, public health infrastructure faces the unprecedented challenge of defending populations that possess highly limited immunity to tropical pathogens like dengue fever and malaria.

Historically, tracking these shifting vectors relied on retrospective medical mapping - which fundamentally relied on documenting active case counts following the onset of an epidemic. This reactive approach is no longer viable in a volatile climate. Modern preventative medicine requires predictive foresight, a capability unlocked by advanced data science. Machine learning architectures, specifically gradient-boosted trees and deep neural networks, excel at processing massive, non-linear environmental datasets. By feeding these algorithms real-time variables, computer scientists can train models to identify subtle ecological patterns that signal an imminent surge in vector populations before human transmission begins.

Algorithmic Architecture: How Random Forest Forecasts Outbreaks

To effectively map malaria risk in rural

tropics, data scientists rely heavily on ensemble learning techniques, most notably the Random Forest (RF) algorithm. A Random Forest operates by constructing a multitude of individual decision trees during the training phase. A decision tree is a flowchart-like model that splits data step-by-step based on specific conditions to reach a final prediction. Each individual tree analyses a randomised subset of the overall climate and biological data, outputting its own isolated prediction of whether or not a specific rural district will experience a malaria spike. The algorithm then collects these findings, utilising a majority-voting consensus mechanism to deliver a singular, highly accurate final forecast [2].

This ensemble learning design is exactly what makes Random Forests far superior to single decision architectures when in rural settings. Rural tropical zones frequently suffer from fragmented data collection due to underfunded local clinics and intermittent satellite coverage. While a standard linear model or standalone decision tree might completely fail or produce skewed results when encountering missing data points, the structural redundancy of a Random Forest ensures effective performance. By averaging the independent insights of hundreds of trees, the algorithm minimises the risk of overfitting. This allows it to successfully isolate critical climate patterns amongst vast amounts of ecological noise. This 'noise' includes irrelevant background variations, like sudden local pollution spikes or unmeasured environmental factors like localised weather anomalies or random variations in wildlife behaviour, that can distort data.

Data Fusion: Tracking the Vector's Lifecycle

The predictive accuracy of the Random Forest model is significantly influenced by the quality of its inputs, requiring a fusion of macroeconomic climate data and microeconomic biological variables. The algorithm is trained on three primary environmental pillars:

- Land Surface Temperature (LST): mosquitoes require optimal ambient warmth to survive and replicate. When temperatures rise past a specific biological threshold, the incubation period of the *Plasmodium* parasite inside the mosquito drastically accelerates [1].

- Precipitation and Humidity: Increased tropical rainfall generates stagnant pools of water in rural areas, creating perfect breeding grounds for mosquito larvae. At the same time, high humidity extends the lifespan of adult vectors, which explains why mosquitoes are primarily found in warmer and wetter climates.

- Normalised Difference Vegetation Index (NDVI): Satellite data measuring dense forest canopies and rural vegetation density helps the algorithm identify hidden pockets of moisture where the vector population thrive outside of standard wetlands.

By synthesising these real time satellite metrics alongside historical localised outbreak data, the Random Forest model computes feature importance rankings. This reveals exactly which environmental trigger (e.g. specific two-week lag in precipitation after an intense heatwave) is the primary driver of an impending epidemic.

Clinical Deployment: Transforming Code into Proactive Medicine

The true value of a Random Forest predictive model lies in its ability to transform raw, environmental data into actionable preventative medicine. This is especially vital for tropical regions with limited resources, such as LICs, NEEs, and areas with a lower development index. Historically, rural healthcare infrastructures in sub-Saharan Africa and Southeast Asia have operated on a reactive basis, deploying medical interventions only after local clinics register a dramatic increase in malaria cases. At this point, local

pharmaceutical supply chains are often overwhelmed, leading to critical shortages of rapid diagnostic tests (RDTs) and artemisinin-based combination therapies (ACTs). RDTs are simple, rapid blood tests used to confirm malaria infections on-site, whereas ACTs serve as the primary, fast-acting medication used to treat confirmed cases. Together, these tools are vital for malaria management; shortages directly lead to preventable deaths and the misuse of alternative treatment.

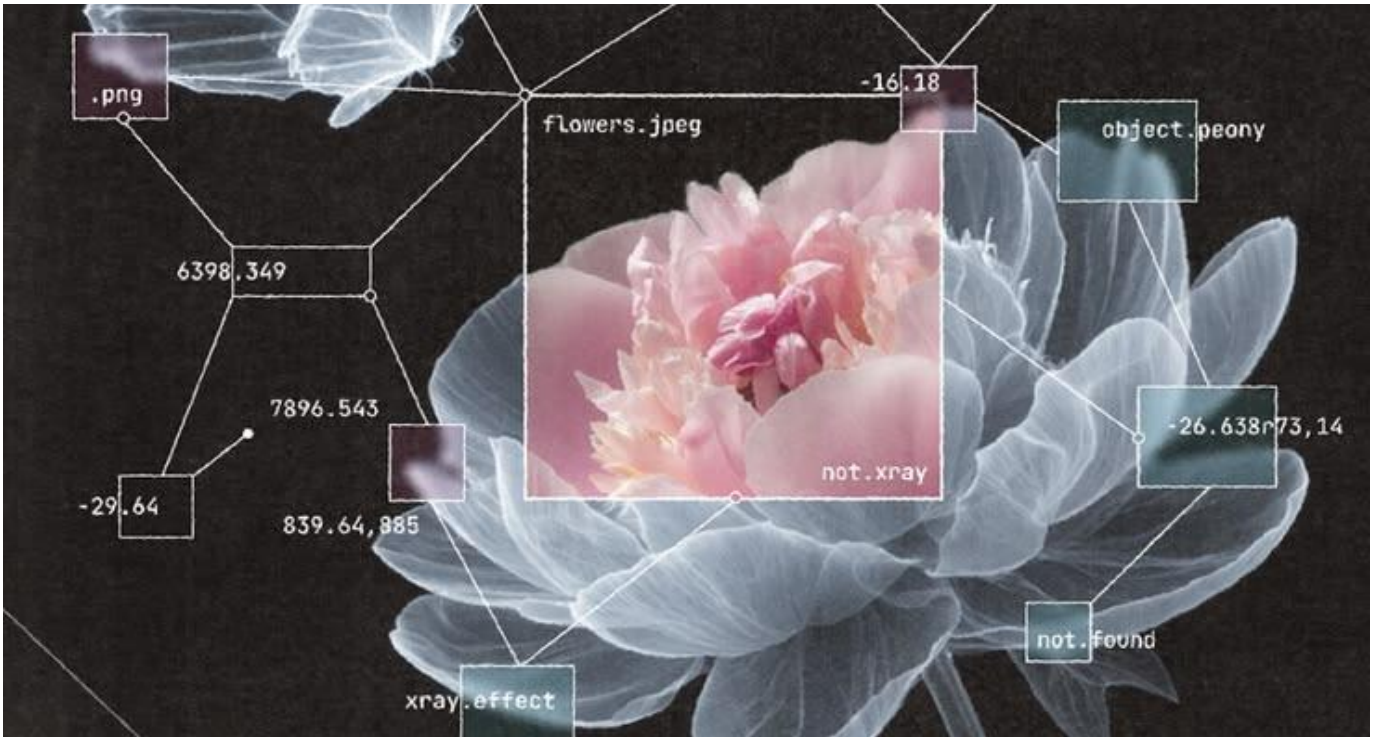
By embedding machine learning models into public health logistics, ministries of health can transition to a proactive paradigm. When the algorithm flags an imminent outbreak hotspot (often 3 to 6 weeks before human transmission peaks), healthcare workers deploy targeted preventative measures. Medical supply chains can pre-position antimalarial medications and diagnostic kits in vulnerable villages before seasonal rains isolate rural roads. Non-governmental organisations (NGOs) can optimise the distribution of long-lasting insecticidal nets (LLINs) to the exact communities identified as high risk, rather than relying on inefficient, blanket distributions [3]. Finally, local vector control teams can execute precise larviciding operations, treating stagnant water bodies with biological agents to kill mosquito larvae before they mature into vectors. This targeted deployment ensures that finite medical and financial resources are focused precisely where and when they will save the most lives. This precision also brings massive economic benefits. Replacing broad, blanket distributions with AI-driven targeting cuts supply chain waste and slash logistical costs. Ultimately, stopping outbreaks earlier lowers healthcare costs and protects local economies by keeping communities healthy and productive.

Conclusion

To conclude, the escalating climate crisis demands a fundamental evolution in how humanity combats infectious diseases. As shifting temperatures and unpredictable precipitation patterns redefine the global boundaries of malaria transmission, traditional, reactive epidemiological frameworks are no longer sufficient to protect vulnerable populations in the rural tropics. The convergence of machine learning and preventative medicine offers a powerful, data-driven path forward. By utilising the robust ensemble architecture of Random Forest algorithms, data scientists and clinicians can decode complex ecological interactions, transforming satellite data into an early warning shield for remote communities, ultimately bridging the gap between computer science, biology and healthcare logistics, not merely optimise public health resources, but also establishes a vital technological barrier capable of safeguarding human life on a rapidly warming planet.

By Kaifa Azad

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Biomimicry: How Nature-Inspired Computer Science Could Change the Climate

Nature: Computer Science's Hidden Gem

Despite the billions of years' worth of evolution in nature, due to human-caused climate change, it has been painted as compromised and vulnerable. This can be seen in the rhetoric of recent papers, presenting nature as "fragile" and "sensitive" [1]. As a result, nature has been undervalued, and its potential undermined. Biomimicry is a field that aims to revolutionise design by integrating the most useful biological structures, countering the vulnerable narrative [2], [3]. This article examines the influence of biological mechanisms through neural networks and membrane systems, and addresses artificial intelligence and membrane gas separation. In addition to this, it analyses environmental impact and mitigation strategies.

Nature Taught Computers to Process

A neural network is an example of a system which has been designed with natural inspirations in mind: a process known as biomimicry. Modelled after nerve cell mechanisms, a neural network contains a congregation of linked nodes [4].

Neurons send electrical signals which are impulses [5], known as "spikes". The signal then travels across the synapse, which can strengthen with more added [6]. The neuron then fires an impulse, which is consequently transmitted to neighbouring neurons by travelling through the cell's axon [5], [6].

When designing neural networks, computer scientists transferred core biological concepts to recreate the human mind: nodes, which represent the neuron; and weights, which represent synapses [4]. In development, this allowed researchers to advance past pure replication: they analysed these most efficient attributes of the brain and recreated them.

Success of neural networks, or ANNs, has been noted in pattern recognition. This has initiated potential for uses in climate research over the coming years, such as provisional weather simulations [7]. Although, there are still setbacks. This includes a lack of research on the topic [8], and worries about impact on the environment, as neural networks require large amounts of energy to perform [9].

Most importantly, ANNs are now being used to make short-term climate predictions with climate change in mind, becoming a step in helping scientists understand how human

impact is affecting the world [7]. Research by [10] found an artificial neural networks model with 1 hidden layer and 10 neurons resulted in 87.945% overall accuracy when supplied with climate change data consisting of maximum and minimum temperature over specific time periods. This could help identify causes and trends in the complex problem of climate change to better understand global warming [10].

Just two years ago another model was developed by [11] within the climate engineering field. While [10] predicts future climate changes, [11] aims to find the forcing behind it. This model was trained on surface temperature responses and image data to detect the causes of temperature changes, achieving an average 0.85 R^2. This could potentially be used to accurately attribute causes to changes in the climate, including anthropogenic effects. Comparing radiative forcing data to external emissions data, this could be used hereafter to point to specific industries or countries with the greatest environmental impact [11].

Influence by the function of brain neurons in computing has shown to be advantageous in climate science and engineering. However, there are still some setbacks: increased energy demand leads to an effect on the environment. As a result, the same technology giving us information about the world has a negative impact on it [9].

Nature Taught Membranes to Communicate

Developed by Gheorghe Păun, membrane computing integrates efficient features from the structures of biological cells as a sector of natural computing to result in a membrane system, or P system, named after Păun [12], [13].

A P system is a computing model made up of a hierarchical structure of membranes in a linked arrangement, where only the objects within them move according to specific rules [12], [13].

The structure of a membrane system is directly inspired by nature: biological cells. Communication is made efficient through the symport and antiport natural operations, which is made even easier when membranes are tightly packed [14].

The system expanded its capabilities by taking inspiration from biology, with incremental improvements eventually leading to further developments. The original single cell influenced system developed into three main models, including the cell-like P systems shaped by the “eukaryotic cell” [13]. This reflects how deeply natural ideas moulded the biomimetic design of P systems, detailing down to even methods of communication.

With some membrane systems having the power to solve computationally difficult problems [13], an application with real-world impact is ecosystem modelling. This is a method of studying ecological populations through stimulating ecosystems to understand their behaviour. Due to the modularity and parallel processing ability of membrane systems, many different environments can be simulated at once [15]. According to research by [15], when inputting environmental data such as reproductive and mortality rate, the error rate stayed below 10% for most years when modelling giant pandas. This reveals the model could be used to track, predict, and understand patterns in the fluctuations in animal populations, including endangered species.

The influence of membranes expands beyond computer science. Technological bio-inspired membranes are an improved approach to carbon capture, inspired by biological processes of active transport and gas exchange [16]. This has resulted in greater gas separation efficiency leading to improved sustainability in reducing CO_2 in the atmosphere: the very same gas that is worsening global warming [16].

The success of P systems demonstrate the technical potential of valuing nature’s intelligence, which can be utilised from

computer science to engineering in order to better understand the changes happening to the world.

Nature's Technological Impact

Over time, nature has developed highly specialised methods that biomimicry transfers into invention. By shifting the framing of nature in research, scientists could transfer fundamental evolution elements into mitigation strategies.

Biomimetic design has produced technologies that transform the way we perceive and understand our environment. From neuron-fuelled climate prediction [10] to membrane-influenced carbon capture [16], nature provides structure for researchers to understand and stop human-caused negative climatal impacts.

Even artificial intelligence, or AI, was developed inspired by human thinking. Although AI is now being used to widen the understanding of climate change we have now, improvement means energy demand from training data [17]. This results in its own negative impacts on the environment. This could be challenged following an idea raised by [18]: taking inspiration from homeostasis to limit the impact of AI on the climate.

Therefore, nature should not be treated as compromised and vulnerable. By taking advantage of its billions of years worth of development and improvement, scientists can further understand and lessen human impact on the climate.

Whereas it may appear the future of technology is against the natural world, learning from what has been here longer than the human race may be the key to understanding it.

By Isobel Parry

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The Fuel Before the Fossil: Solving Climate Change With the Reaction That Started It

The word “fossil” comes from Latin *fossilis* meaning “dug up.” Coal, oil, and natural gas earn that name honestly: they are the million-year-old remains of organisms that once used sunlight and photosynthesis to store carbon-based energy. When fossil fuels are burned, that long-stored energy is released, along with carbon dioxide. By 2025, fossil fuels caused global carbon dioxide emissions to hit a record 38.1 gigatons, making up around 74% of all global greenhouse gases [1], [2]. This is part of why burning fossil fuels is the primary contributor to climate change, pushing global temperatures to limits scientists believe can become irreversible. Any viable alternative must replicate the energy-dense storage capabilities of fossil fuels, while reaching the scale necessary to meet global dependence and industrial demand.

Limitations of Current Options

Solar panels are heavily relied upon as the default alternative, but they generate electricity rather than fuel. This distinction matters most in sectors such as aviation, cargo shipping, and heavy manufacturing, which depend on dense, storable chemical fuel rather than electricity. The standard power-producing option is electrolysis, which uses electricity to split water into hydrogen. At an industrial scale, however, electrolysis is highly energy-intensive, and the electricity it requires is typically made by burning fossil fuels. Because of these limitations, scientists are seeking a solution that skips this electrical step completely and uses sunlight to create a fuel molecule directly. This approach is called artificial photosynthesis, and it takes two main forms: splitting water to produce hydrogen fuel, or catching carbon dioxide directly and chemically converting it into liquid fuels such as methanol, mirroring the way natural photosynthesis fixes carbon into sugars. The most promising solution, however, is rooted in the very biological process that created fossil fuels in the first place.

An Unmatched Catalyst

For decades, scientists have attempted to design a catalyst capable of splitting water into hydrogen and oxygen using only sunlight, at room temperature, without rare or toxic materials, but none have fully succeeded. Nature, however, appears to have solved this problem first. Through billions of years of evolution, a natural catalyst that meets all of the requirements is located inside a plant protein complex called Photosystem II (PSII), which is a cluster of four manganese atoms, one calcium atom, and five oxygen atoms. This is known as the Mn_4CaO_5 cluster or oxygen evolving complex (OEC) [3]. A 2025 review comparing engineered water splitting catalysts to the cluster found that artificial versions exhibit poorer long-term stability and an inability to recover their activity post-damage, as well as a continued reliance on costly components [4]. The cluster completed the same reaction using common elements from the Earth’s crust, at an ordinary temperature and pressure.

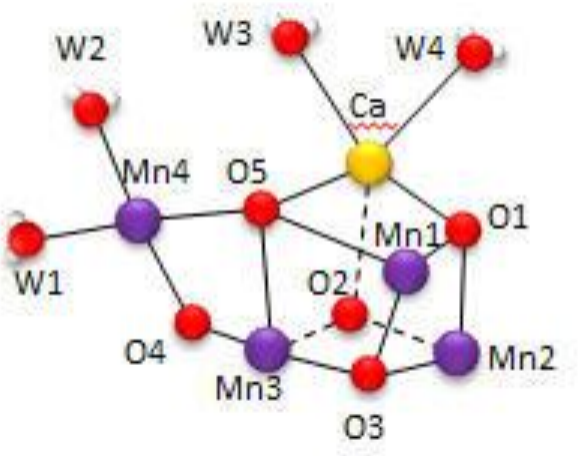


Fig. 1. Visualization of Mn_4CaO_5 cluster’s structure at the core of PSII’s OEC [9].

Splitting water is a complex process that is far from instantaneous. Each photon that strikes PSII removes a single electron from the manganese cluster, an amount insufficient on its own to split water. The cluster instead accumulates four separate bursts of solar energy in sequence, a process known as the Kok cycle: with each photon absorbed, the cluster advances through one of 5 oxidation states, labeled S_0 through S_4, storing more chemical potential at each step. Only once it reaches the final state does it hold enough oxidizing power to break apart two water molecules simultaneously and release one molecule of oxygen. Recent crystallography studies, using fast x-ray pulses to capture atomic motion in real time, show a new water molecule binding to a specific manganese site before this final step, a step that was previously unknown to scientists just a few years prior [3]. These discoveries are crucial because any attempts to create a working copy of this reaction must start with a deep understanding of exactly how the original works. What looks like an ordinary step of photosynthesis is, at the molecular level, one of nature's most sophisticated catalytic processes that no current system has successfully replicated.

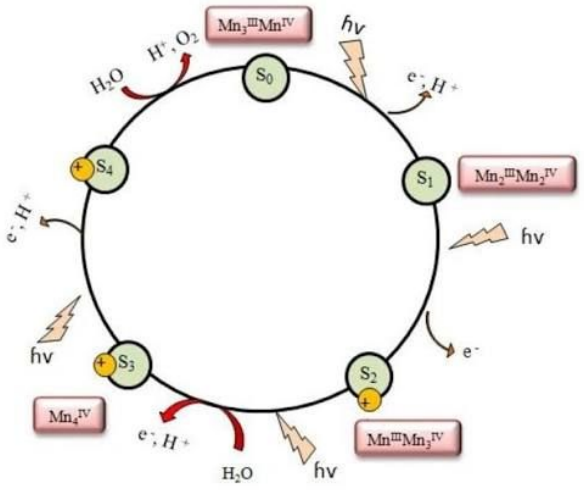


Fig. 2. Kok cycle showing the 5 oxidation states (S_0-S_4) the cluster passes as it accumulates the needed 4 photons before releasing oxygen [10].

The advantage of the cluster is much more than just its chemistry. Splitting water is oxidatively demanding, and its surrounding reaction center takes on damage even under normal light conditions. Instead of degrading permanently, the plant breaks down the damaged protein at the reaction center and rebuilds the catalytic site from newly made components. This repair cycle can happen as often as twice an hour under high light exposure [4]. Current catalysts have no comparable repair cycle. Once damaged, they gradually decline in performance until they must be replaced entirely.

An Open Question in Physics

The step immediately before this reaction, when sunlight travels to the reaction center of the cluster, is still scientifically unsettled. In 2007, physicists found that this energy transfer could behave like a quantum wave rather than move sequentially from one molecule to the next, as had previously been assumed. This finding was later confirmed to hold even at biological temperatures [5], [6]. Quantum coherence, in this context, refers to the possibility that a photon's energy travels toward the reaction center as a wave sampling multiple molecular pathways at once, rather than jumping between individual molecules one at a time. Other studies have countered that the observed effect may instead result from vibrational motion within the molecules rather than genuine quantum coherence, so the question remains open as of today [7]. Scientists do not yet fully understand this early stage of photosynthesis, but understanding it is just as important for copying PSII as is its water-splitting chemistry.

The Synthetic Leaf

Every breakthrough in this field comes down to the single goal of replicating the chemistry performed by the Mn_4CaO_5 cluster, whether through fully synthetic materials or through bio-hybrid systems that connect living photosynthetic organisms directly with electrodes. In a bio-hybrid photoanode, whole cyanobacteria or isolated PSII complexes are attached to an electrode surface, allowing the electrons made by natural

water-splitting to be captured and used directly, rather than requiring the entire catalytic system to be re-engineered from scratch [4]. In 2025, Ulsan National Institute of Science and Technology (NIST) researchers built an artificial leaf that splits water using just sunlight. It reached a solar to hydrogen conversion efficiency of 11.2%, which crossed the threshold widely considered necessary to make the device commercially practical [8]. This performance was maintained even after 140 hours of continuous use while keeping 99% of its initial efficiency [8]. It was later scaled to a 16-square-centimeter photoelectrochemical cell, a device that uses semiconductor electrodes, called photoanodes and photocathodes, to absorb sunlight and directly drive the water-splitting reaction at their surfaces. This made it the largest device to date to clear the 10% mark with stability [8].

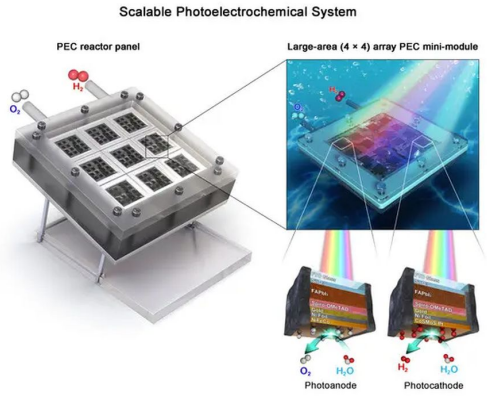


Fig. 3. UNIST water splitting system made by a 4 by 4 array of photoanode and photocathode cells [8].

Instead of relying on rare and costly metals like platinum and iridium, researchers used a mix of nickel, iron, and cobalt catalysts along with a chlorine-treated-perovskite light absorber [8]. This indicates the same shift towards the common, abundant materials the Mn_4CaO_5 cluster achieved, billions of years earlier. Reaching this level of efficiency, durability, and scale is one of the strongest signs yet that artificial photosynthesis has the ability to move beyond laboratory chemistry and towards industrial energy production, allowing it to finally replace fossil fuels in areas that electricity alone cannot. Making hydrogen this way costs more than conventional methods and the synthetic catalysts used still remain less efficient and stable than the cluster. However, every new finding and study is surely but steadily closing the gap.

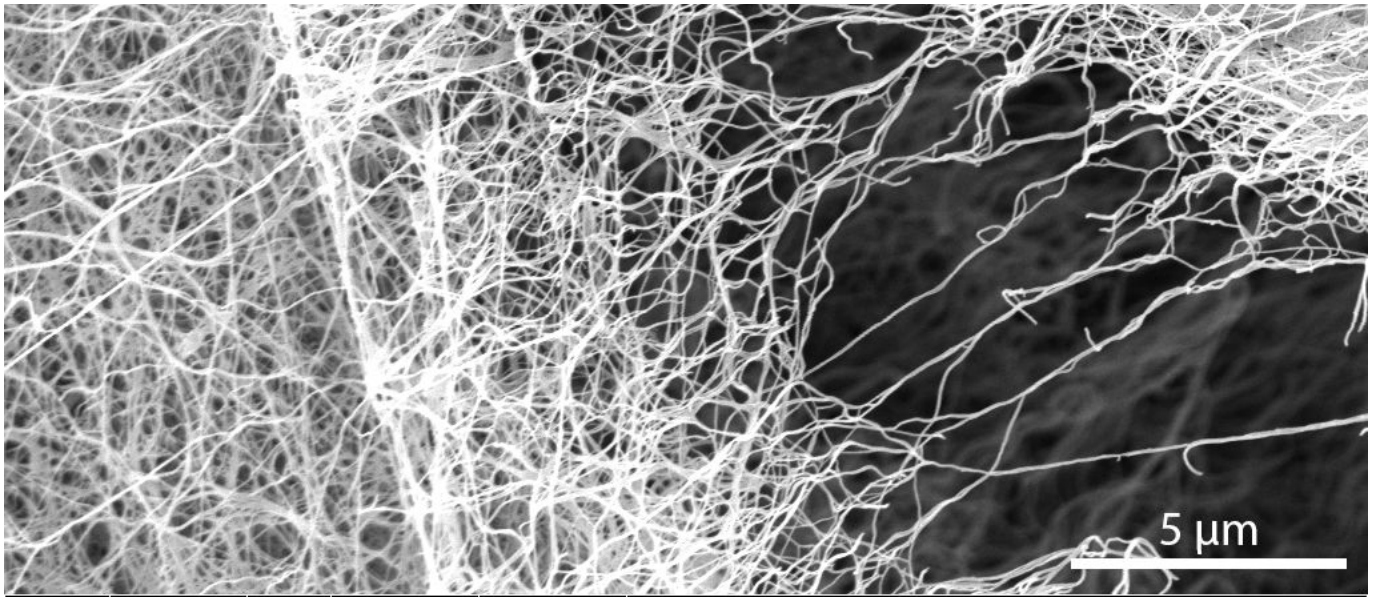
Conclusion

The paradox of the ongoing energy crisis is that fossil fuels and their replacement have the same origin. With photosynthesis, organisms spend millions of years turning sunlight into stable carbon-based-fuel, but that energy has rapidly been released within just a few centuries by the burning of fossil fuels. If a single natural reaction is responsible for storing the fuel humanity now depends on, a faster version of that same reaction is a reasonable place to look for what could replace it. The Mn_4CaO_5 cluster is the closest known model that makes fuel exactly the way fossil fuels were originally made. Climate change now does not have to be treated as a problem that needs a brand new invention to be solved. Photosynthesis is proof that the technology best suited to replace fossil fuels already exists; what remains is understanding it well enough to copy.

By Sragvi Madishetty

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Nature's Environmental Micro-Bioengineers: Sustainable Biomaterials Derived from Bacterial Cellulose

With growing concerns about synthetic plastic waste in the environment, materials scientists are increasingly prioritizing microbial products as a sustainable source of biomaterials. Specifically, the protective extracellular matrix produced by certain bacteria offers a compelling alternative to petroleum-based materials. This material, known as bacterial cellulose (BC), is an extracellular polysaccharide with high tensile strength, purity, and crystallinity, making it a promising biomaterial. Its inherent biodegradability and biocompatibility promote environmental protection, resource efficiency, and integration into the circular bioeconomy [1]. Scaling BC towards industrial production could be a key factor for replacing synthetic polymers and advancing sustainable materials.



Fig. 1. Bacterial cellulose biosynthesized by Starkeya sp. [1]

Synthesizing Bacterial Cellulose

Bacterial cellulose (BC) is a homopolymer composed of chains of glucose molecules, sharing the same molecular formula as plant cellulose, but free from impurities such as lignin and hemicellulose. It is produced by a wide range of bacterial genera, including *Komagataeibacter, Aerobacter, Achromobacter, Azotobacter, Rhizobium, Agrobacterium, Rhodobacter*, and *Sarcina*. Most reports on bacterial cellulose production focus on *Komagataeibacter* as a model organism because of its high cellulose yield [2].

The chemical structure of BC is made up of linear chains of β-D-glucopyranose units joined via β-1,4-glycosidic linkages. BC production depends on carbon sources such as glucose, fructose, sucrose, and glycerol, which are converted into metabolic intermediates, yielding uridine diphosphoglucose (UDP-glucose). The synthesis is carried out by membrane-integrated cellulose synthase complexes (CSCs) encoded by the *bcsABCD* operons. CSCs use UDP-glucose as a substrate and polymerize it into a long chain of β-1,4-glucan. These chains are then transported out of the pores of its outer cell membrane, forming a 3D mesh of microfibrils [1], [2].

Optimization Through Agro-Industrial Waste

The efficiency of BC biosynthesis relies

heavily on the tailored balance of the medium composition, including carbon substrates (glucose, fructose, and sucrose), nitrogen donors (yeast extract, peptone, and casein), biochemical stimulants (ethanol and glycerol), and trace minerals and vitamins. Modulating these nutritional balances affects adenosine triphosphate (ATP) generation and the availability of UDP-glucose precursors. However, the obstacle in optimizing BC growth and mass production often stems from the prohibitive cost of utilizing analytical reagents, which could hinder further expansion into industrial products and commercialization [2].

Recent studies are investigating alternative carbon sources that could lower production costs, such as using agro-industrial waste and biodegradable waste for BC synthesis. In particular, fruit and vegetable residues undergo mechanical and enzymatic processes that hydrolyze their polysaccharides into a range of monosaccharides and disaccharides, which are then used for microbial growth and cellulose production. This valorization strategy not only reduces carbon source costs but also promotes the recycling of biodegradable waste into an abundant, carbon-based, fermentable medium. For instance, pineapple peel waste was extensively used and evaluated as a substrate for *Komagataeibacter saccharivorans* and other acetic acid bacteria. This substrate proved to be an effective carbon source that significantly enhanced BC productivity when analyzed using Response Surface Methodology (RSM) and Artificial Neural Networks (ANNs) [3], [4]. Similarly, mango pulp was investigated, yielding 6.32 g/L of BC, demonstrating potential for biomaterial production [5]. Furthermore, rice straw hydrolysate was examined for the cost-effective production of BC, with a reported yield of 7.17 ± 0.05 g/L, higher than that of BC produced from various lignocellulosic wastes [6].

Beyond the alternative substrates, investigating varying microbial strains and co-cultivation is essential for optimizing BC synthesis from waste media. A novel lactic acid bacterium, *Levilactobacillus brevis*, isolated from nata de coco wastewater, was evaluated for its ability to bioconvert an agro-industrial substrate into BC using Response Surface Methodology (RSM). This approach not only improves BC productivity but also enhances its physicochemical properties, such as tensile strength, crystallinity, and purity, thereby meeting the standards for commercial biomedical and packaging applications [7].

Studies that offer various approaches to utilizing alternative substrates and further scrutinize microbes under varying conditions for optimization and quality enhancement are a pivotal advancement in BC biotechnology, expanding opportunities and advantages as a sustainable biomaterial. What should be fully addressed, based on these advancements, is their current applications across industries and their adoption as commercial products.

Application and Climate Impact

With over 460 million tonnes of plastic produced and 20 million metric tonnes of plastic litter discarded in the environment annually, plastic waste continues to intensify environmental degradation, leading to biodiversity loss and threatening human health [8]. Common thermoplastic polymers used in industrial production, such as polyethylene terephthalate (PET), contribute significantly to environmental persistence and chemical toxicity risks. It is estimated that 2,100 species, including endangered species, are affected by plastic pollution annually, resulting in mortality from ingestion and entanglement [9]. These pollutants also cause adverse environmental effects through bioaccumulation in the food chain, atmospheric transport, and soil leaching. Under natural forces such as solar radiation, wind, and currents, plastics fragment into micro- and nanoplastic particles; because

of their size, they readily cross cell membranes and accumulate in living tissue. Concurrently, these degrading plastics release toxic compounds, including bisphenol A (BPA) and heavy metals, that are associated with respiratory illness and cancer [10]-[12].

To address this concern, materials science has shifted towards using biodegradable microbial products, which are integrated into various applications across different fields. Various studies have analyzed applications of BC, ranging from biomedical uses to advances in its use as a green technology. The aforementioned studies, including those that use agro-industrial waste as an alternative substrate, have examined BC as a potential biomaterial. This material exhibits high tensile strength, full biodegradability, excellent biocompatibility, and high purity [13], [14].

Recent studies investigating the refinement of BC's physicochemical characteristics using agro-industrial waste highlight diverse environmental sustainability benefits. By valorizing biodegradable waste, BC can potentially replace petroleum-derived plastics, promoting resource efficiency and supporting environmental protection [13].

Although integrating BC into industrial-scale production can be challenging, cultivation methods may hinder this expansion. Bacteria used in BC production are aerobic strains that rely heavily on a continuous supply of oxygen. Conventionally, methods such as static cultivation are used to achieve high crystallinity and tensile strength; however, in this method, oxygen can enter the system only at the air-liquid interface via passive diffusion. Eventually, the dense and thick gelatinous BC pellicle impedes oxygen supply to aerobic bacteria embedded deep within the medium. This hinders metabolic activity and cellulose synthesis capacity, resulting in lower volumetric production [15].

Current research is investigating strategies such as using agro-industrial waste for cheaper production, co-fermentation systems, genetic engineering, and pulse-feeding systems. Continued advances in these studies could be a driving force in the biopolymer production sector [13].

Future Development

The integration of BC within the circular bioeconomy is expanding across biomedicine and advancing towards the industrial scale. Due to its nanostructured fibers, which exhibit high crystallinity and tensile strength, BC has potential as a material for 3D tissue-engineering scaffolds. Additionally, BC's biocompatibility and water-holding characteristics can be utilized for hydrogel-based transdermal drug delivery, while its exceptional strength and biodegradability make it a candidate for smart food packaging [16], [17].

The large-scale application of BC relies on replacing reagent-grade carbon with material derived from the valorization of agro-industrial waste. This promotes the recycling of biodegradable waste as fermentation feedstocks, reduces production costs, and addresses waste management. Future development could leverage artificial intelligence (AI) and predictive machine learning models to optimize fermentation parameters, thereby calibrating variables such as pH, temperature, and aeration to maximize BC yield and quality [18]. Genetic engineering and mutagenesis techniques have also generated hyperproducing strains that can withstand harsh conditions when using waste-derived feedstock [19].

Conclusion

On a commercial scale, where plastic is widely used in the production of goods, a shift towards advancing sustainable biomaterials is imperative. Bacterial cellulose, as a sustainable driver for the circular bioeconomy, represents a transformative biotechnology-based solution.

By deciphering metabolic pathways, applying genetic engineering, and using agro-industrial waste and alternative substrates, future research can translate this biopolymer concept into a commercial application by reducing production costs and achieving sustainable industrial manufacturing. It is also essential in reducing reliance on petroleum-based materials, lowering the carbon footprint, and fostering environmental sustainability.

By Michaella Dorothy D. Babac

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Mycorrhizal Fungi: An Overlooked Organism in Climate Change Mitigation

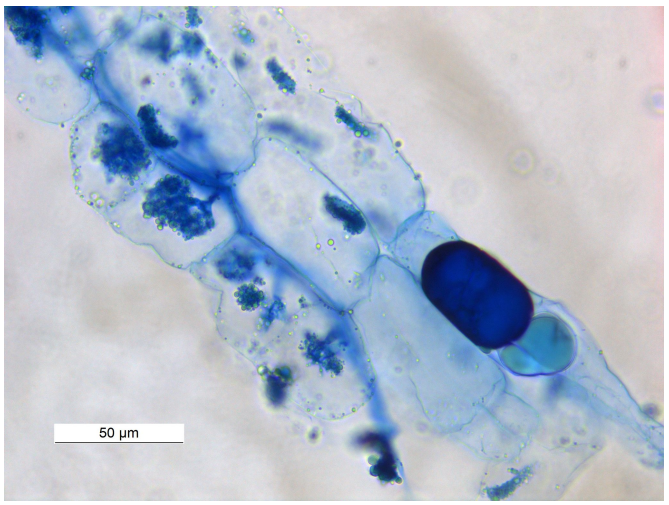


Fig. 1. Vesicular Arbuscular Mycorrhizae [1]

It constitutes no great revelation that the climate crisis has grown out of control in recent years. With temperatures rising annually, the resulting unpredictable weather, biodiversity loss, and wildfire poverty cannot be ignored. One of the main contributors to the rise in atmospheric greenhouse gases is carbon dioxide (CO_2), which is primarily driven by human activities such as the burning of fossil fuels. As carbon dioxide accumulates, it absorbs heat radiating from the Earth and traps it within the atmosphere raising temperatures. Reaching a global high of the atmosphere in 2024 at 422.7 parts per million (ppm), the world is only getting hotter [2].

Through biological sequestration, mycorrhizal fungi, a species lying beneath the surface of the earth, may be an overlooked component to the battle against climate change.
While the fungi are a microscopic organism, their symbiotic relationship with their plant hosts influences many aspects related to its surrounding climate. This includes moving and stabilizing fixed carbon in soil, improving overall ecosystem resilience against climate stress, and stabilizing soil structure [3]. Correspondingly, the conservation of the mycorrhizae becomes notable in climate mitigation as well as adaptation.

The Plant-Fungi Relationship

The mycorrhizal fungi is a microscopic species of fungi that closely associates with plant roots–specifically those of the vascular family. The word “mycorrhiza” comes from the Greek language, “myco” meaning fungus, and “rhiza” for root. Together, it is the name for the symbiotic existence of the fungi and its plant host in which mutually benefit off of resources that they trade with each other [4], [5]. With global topsoils holding roughly 110 quadrillion kilometers of mycorrhizal fungal network, the extensive species forms interdependent relationships with 70% of plant species globally [6].

Primarily recognized for absorbing carbon dioxide, plants use photosynthesis to absorb CO_2 from the atmosphere, converting it into glucose compounds for energy [7]. While some of the fixed CO_2 goes to the host plant, the rest goes through the plant roots to the fungi dwelling below the soil [8], [9]. In exchange for the photosynthetically derived carbohydrates and fats, the mycorrhizae utilize their extensive network of thread-like hyphae to absorb an deliver essential nutrients and minerals—such as water, phosphorus, nitrogen, and sulfur—that the plant needs to survive and thrive [9], [10].

Mycorrhizae and Carbon Sequestration

From the fixed carbon transported to the fungal hyphae, the mycorrhizae uses it to build its own biomass, constructing hyphae and supporting its mycelium network. The fungi produces compounds such as exudates, chitin, and melanin, which are immobilized by soil microbes keeping carbon in the soil in the form of “carbon pools” [9]. The connection the carbon possesses with the soil protects the organism from microbial deterioration, heightening temporary terrestrial storage for carbon [9].

When the mycorrhizal biomass eventually dies it becomes known as necromass which stores carbon by acting as a physical support for soil. The necromass acts as a

framework of organic material that soil particles attach to, facilitating in increasing soil aggregate size. The soil aggregates then protect the necromass from further decomposition, slowing the release of carbon back into the atmosphere [9]. However when soil is disturbed, microbial activity declines and soil aggregates break down returning CO_2 to the atmosphere resetting the cycle [12]. This indicates that the fungi doesn't *trap* fixed carbon, but rather works with their plant counterparts to transport and temporarily store CO_2 [11].

Strengthening Ecosystem Resilience

Climate change not only results in the increase of atmospheric CO_2, but alters the environment that the mycorrhizae live in [13]. In addition to carbon sequestration, through the symbiotic nutrient cycling that takes place between the host plant and the fungi, the mycorrhizae also increases ecosystem resilience by providing the host with the necessary nutrients to survive in extreme weather events and stress factors attributed to climate change [14].

Droughts represent one of the most extreme weather events driven by climate change. As temperatures rise globally, water scarcity intensifies, becoming more frequent and severe [13]. During the crises, the mycorrhizae act as an extension of plants' roots, allowing the host to reach more water in the soil that might otherwise be out of reach for the plants [15]. As more regular and acute droughts reduce the productivity and survival of plants [13], the mycorrhizae, by improving access to water and nutrients, helps to increase tolerance and sustain plant productivity under climate stress.

Similarly, soil salinity has also been significantly influenced by global warming. Climate change dramatically alters environments and accelerates soil salinization through rising sea levels, higher temperatures, and irregular precipitation [16]. Arbuscular mycorrhizal fungi, a form of mycorrhizae, has been demonstrated to improve salinity tolerance with nutrient acquisition [3], altering osmotic balance and carbohydrates [17], and improving water absorption [15]. In altering plant structure, the mycorrhizae improves plant resilience, allowing species to better thrive under high salinity. Subsequently it supports wider ecosystem stability through sustaining food production enhancing agriculture cultivation in high saline soils [18].

By maintaining plant productivity under climate stressful conditions, the fungi indirectly contributes to carbon uptake and climate mitigation.

Stabilization of Soil and Carbon Storage

Shifts in climate patterns directly impact soil stability and vulnerability, altering the capacity of soil-carbon sequestration [19]. The mycorrhizae hyphae physically stabilizes soil structure through mycelium entanglement (physical means). The network combined with plant roots, direct the formation of microaggregates, curbing erosion [20], [12]. The hyphae and roots bind the microaggregates into macroaggregates [20], which are significant indicators of soil health by enhancing porosity, protecting organic matter, and improving plant root growth. Accordingly, by improving soil aggregation and porosity the mycorrhizae creates healthy soil that is necessary for maintaining CO_2 in terrestrial systems and preventing it from returning to the air [21].

Threats to Mycorrhizae Networks

As the mycorrhizae has a monumental influence in ecosystem resilience and carbon storage, the destruction of their habitat can consequently hinder its positive effects [22].

One of the major threats to mycorrhizae is the use of chemicals and pesticides in the soil. The overabundance of fertilizer and chemical use causes nutrient imbalances, often increasing phosphorus levels [22], [27]. The phosphorus along with other

excessive nutrients in the soil reduce the host plant's reliance on their symbiotic fungi partners—no longer requiring fungal support for nutrient synthesis and absorption. This disruption of symbiosis often indirectly kills mycorrhizae by slowing or stopping the flow of essential sugar from the plant's roots which it requires to thrive [22], [27].

Similarly, soil changes caused by anthropogenic disturbances such as urbanization, and deforestation, lead to soil compaction and erosion which directly reduces fungal habitats and their ability to create mycelial networks. With actions such as extensive tillage, aggregates are further broken down, becoming incapable to proceed as carbon storages [12].

Conservation through Technology

As technology advances ever increasingly, it offers innovative solutions for conserving mycorrhizae fungi.
Although the unseen species spans quadrillions of kilometers, fewer than 90% of hotspots lie within conservation areas [23]. To address this issue, SPUN, the Society for the Protection of Underground Networks, made a breakthrough in technology and science in June 2026, developing a digital model mapping out the microscopic species [24]. The Underground Atlas, as the name suggests, serves as a potential pointer towards climate change strategies, land management, legal, and restorative action through indicating potential threats which would otherwise remain unseen [23].

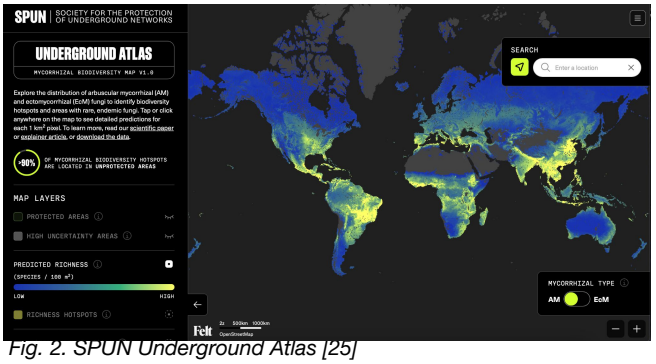


Fig. 2. SPUN Underground Atlas [25]

Beyond mapping hidden ecosystems, the future of mycorrhizae preservation lies with the convergence of machine learning models with fungal cultivation, and molecular engineering. Through modern biotechnology, machine learning models are proposed to design gene edited fungi more efficiently. As co-culture techniques advance, this optimizes metabolic engineering, expanding the possibilities of genomic manipulation using CRISPR and RNA interference (RNAi). The genetically edited fungi suggest enhanced traits, such as increased production of biofuel, enzymes, and pharmaceuticals—tasks which previously conventional *in vitro* culture systems struggled to achieve as the fungi were incapable of completing their life cycle or growing independently in the lab. In addition to predicting modified gene outcomes, as well as discovering new fungal networks, the application of biotechnology is suggested to reduce excessive trials and costs, potentially developing more efficient and sustainable mycorrhizae systems [28].

Conclusion

As a significant part of biological carbon sequestration, soil stability, and climate resilience, the mycorrhizae has a noteworthy impact on the surrounding terrestrial environment. While the mycorrhizae is not a direct solution to solving the climate crisis, the symbiotic relationship they possess with their plant hosts demonstrates how strategies can be looked for not only above ground, but below ground. Only recently realized as a "globally significant" to climate change mitigation in 2023 [26], the mycorrhizae poses as a notable asset to the climate—so long as efforts are continued to be put into its conservation.

By Jaymie Wang

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Biomimetic infrastructure introduced through the Living Building Challenge; advocacy for an ecologically healthy and environmentally net positive architectural future

What would the architectural anatomy of a flower feel like? The softness of the stigma offers shelter while the stamen releases an aromatic freshness that engages the olfactory senses. In winter, the flower petals contract to shield the core from harsh elements; in summer, they unfurl to absorb solar radiation for photosynthesis [1], optimizing energy allocation towards growth and seed maturation. These efficient biological processes of the flora are manifested in the Living Building Challenge (LBC), created by the Living Future Organization in 2006; it is recognized as one of the most rigorous green building certifications [2], providing a framework that approaches sustainability through 7 performance areas which oversee 20 imperatives, known as Petals: Place, Water, Energy, Health & Happiness, Materials, Equity, and Beauty.

The LBC drives adaptive innovation to yield net positive environmental impacts and interweave nature into the cultural fabric of communities. It seeks to fundamentally reshape how each act of design and construction can be reiterated to positively impact the greater community; it satisfies the brain's desire for order through quantitative goals as well as intrinsic human relationship with nature [3], operating as a philosophy first and a certification second. The clear and evident framework highlights the symbiotic and interdependent relationships between people, community and nature as the world becomes increasingly urbanized and human connection to nature is severed.

The seven petals address seven key aspects of sustainability within architecture. They are divided into 20 specific imperatives, and once a building has become certified in all seven petals, it becomes a living building. The petals primary aim is to foster community through nature while outlining goals for positive environmental impact in a cohesive framework.

Prevalent modernized urban architecture diminishes nature's restorative capacity and depletes human energy. The LBC's strategy to this change is to adopt frameworks that prioritize human health and wellbeing through its Health and Happiness, Beauty, and Equity Petals. Observing the rich history of architecture, evident patterns reveal how these spaces were designed to alter human consciousness through space, whether that be through geometry, acoustics or light. Certain factors induce different neurological and affective

responses—specialized cells in the human brains shape the behaviors and cognitive states of those who live within the space [4]. Much like how a flower uses its petals to allocate how much light it lets in for its growth, human circadian biology has shown how daylighting stimulates photosensitive retinal cells that regulate cortisol and enhance cognitive focus [5]. Fresh air and natural light is an imperative of the LBC that requires these spaces to provide things such as operable windows and dynamic daylighting, preserving our circadian rhythms . Additionally, targeted exhaust is installed to prevent air recirculation and contaminant barriers are used to reduce indoor air pollution. This PETAL also introduces biophilic design: a concept that refers to how nature triggers parasympathetic nervous system activation within the human body that reduces stress. Linking to the intrinsic human affinity for aesthetic harmony [6]—the Beauty Petal enforces design elements intended to evoke serenity and biophilic satisfaction. To prioritize aesthetics, this PETAL requires design features that are purely intended for human appreciation such as art installations or architectural statements. However, in order to create a truly inclusive space, the LBC has integrated an Equity Petal, designed to foster human connection and decrease car dependency by prioritizing walkability and visual connections to nature [7].

Collectively, these factors are critical in optimizing human health and restoring natural connection. In recent years, sustainable living has been placed on a pedestal, a future standard that's unanimously agreed upon as an imperative. The millennium project, an independent non profit think tank analyzes policy across the globe to project the constantly updated “15 Global Challenges”. The number one challenge humanity currently faces is prioritizing sustainable development while mitigating the impacts of climate change. [9]” While there may be a rise in environmental consciousness, the commodification of finite resources and human health has become increasingly advertised as the ultimate status of wealth and health [8]. Despite how global trends are constantly changing and being swayed by socio-economic policy, the LBC and PETALS still remain incredibly relevant in the fight against climate change through sustainable systems.

Prior to anthropogenic disruption and topsoil degradation, construction sites contained biodiverse ecological systems; consequently, preserving the ecological integrity of the land must be a primary goal in architectural development. To contain the environmental impacts, the Place PETAL sets limits to the growth of a building within the natural habitat. A fundamental ideology of this standard is to not take more than needed. The Limits to Growth Imperative of LBC 3.1 allowed construction only on greyfields or brownfields, and projects were required to restore the natural landscape without petrochemical fertilizers or pesticides[10]. Additionally, all buildings should include on site sustenance in the form of food production. To offset construction impacts, LBC 3.1 requires projects to preserve an equal amount of land in the process of a land trust, such as the habitat exchange program[10]. This minimizes the environmental stress traditional buildings have on their surrounding environment.

Energy conservation sits at the core of the LBC framework. To counter the anthropogenic exhaustion of finite resources, buildings must achieve net positive energy profiles and generate surplus power beyond operational demand. This is reflected in the LBCs, *Water*, *Energy* and *Materials* PETALS. LBC 3.1 has the imperative of a net positive water production, producing more water than it consumes within the limits of local water sources [11]. To restore natural water flows, living buildings are only allowed to utilize stormwater and harvested rainwater, which can be filtered through water resilience systems. Greywater—water free from fecal contamination—is to be directed for

filtration and released back into the soil. Blackwater systems—water systems that handle sewage—are the most difficult to filter and are eliminated through composting toilets. These closed-loop hydrological systems mirror biomimetic constructed wetlands, converting greywater into reusable irrigation effluent. Notwithstanding its benefits, achieving hydrological independence persists as a challenge in arid regions and high density urban environments bound to centralized water networks.

The Energy PETAL facilitates the mitigation of climate change, traditional energy sources such as fossil fuels and coal are the primary contributors to Green-house-gas emissions. To mitigate environmental strain, the energy petal mandates net positive renewable energy generation, requiring buildings to produce at least 105% of their annual energy demand. For most places, solar energy is the most efficient and accessible form of renewable energy, with 900 PV solar panels generating over 400,000 kWh annually, enough to power an entire building's electricity demands. Lastly, the Materials PETAL seeks to endorse products that are biologically safe for all species, regulating the materials used based on environmental impact in order to foster responsible materials economy and promote fair trade. The 5 primary targets include red list-free materials—created in 2006 by the International Living Future Institute to identify the "worst in class" components of construction—embodied carbon tracking was also adopted to mitigate greenhouse gas emissions. Additionally the use of salvaged and responsibly sourced materials are mandatory, with consideration of waste reduction during construction. Enforcing red list compliance removes toxins and endocrine disruptors from the building, as well as sourcing reused materials and recycling 99% of construction waste. Moreover, to maintain responsibility on industrial extraction of natural resources, LBC 3.1 has strict regulations that need to be satisfied, ensuring fair trade of materials [12].

The day to day operations, manufacturing and transport of construction materials for buildings make them accountable for roughly 37% of global carbon emissions. This statistic is a direct result of the heightened demand for comfort within living spaces. However, the LBC proves that it is possible to build spaces that are 100% self-sustaining in terms of energy and water without coming at the expense of comfort, and instead improving general living conditions by reinforcing the connection between humanity and nature. While individual buildings may seemingly have minimal benefits against climate change on a global scale, the hundreds of certified living buildings worldwide demonstrate that the LBC is viable when integrated correctly. When executed effectively, the framework establishes a paradigm of sustainable design that creates environmentally conscious and ecologically self-sufficient spaces. This challenges the conventions of traditional construction by setting every design decision as an opportunity to maximize sustainability. By fulfilling all 20 imperatives an architectural project functions as an engineered mimicry of a natural net-positive ecosystem.

By Meixi Li

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Asphalt – Is the State of the Roads Humanity's Fault?

As the world continues to suffocate, the state of the British weather becomes all the more dire with temperatures as of the 16th of August having reached highs of 38.1 degrees Celsius [1] and multiple heatwaves occurring since the end of May. This is not just an outlying year, however, as records were broken in 2022[1] with a temperature of 40.3 degrees Celsius recorded in England and noticeably warmer summers occurring over the past few years. In addition to the estimated death toll of more than 2700 people this year alone[1], the illusion that the nation is prepared for this fiery future is showing larger cracks every week.

Roads exist as a dull, easily overlooked and underappreciated part of daily life for almost everybody. From commuters to lorry drivers, government numbers show that 342.6 billion miles were driven by vehicles on British roads in 2025[2]. The logical conclusion that many could come to is that each road and lane makes up the system that keeps the heart of Britain beating. Without them, millions of commuters would be unable to access work and food, and other essentials would be stranded at depots around the country, sending life to a grinding halt for almost everyone. It would be reasonable to assume, therefore, that each road is prepared for all kinds of weather: salt in order to counter the ice each winter, drains that remove water from the surface during and after rain. These both exist to keep road users safe during potentially dangerous weather events and conditions. However, the recent warmer weather has left some people wondering if this road network is safe.

Roads begin life in the form of bitumen, one of the heaviest long chain hydrocarbons, the organic compounds between Hydrogen and Carbon that make up oil. After the oil has been cracked, this bitumen is removed, mixed with other chemicals, and heated, becoming the substance which is spread over roads in order to bind all other components in the road surface together. The use of bitumen allows roads to be more durable, reducing the required number of repairs. Unfortunately, due to its origin, the production of asphalt is neither sustainable nor beneficial for the environment as oil is a non-renewable resource, and oil drilling negatively impacts local ecosystems. Furthermore, oil spills can occur, which may lead to the exposure of wildlife, such as birds, to the chemicals. These chemical exposures can cause seabirds to lose the waterproof nature of their feathers, which allows for hypothermia from the cold water to kill them. This not only reduces local biodiversity, but could also lead to an imbalance in the food chain. Fish can also be affected, with oil spills reducing growth, eroding fins, and impairing reproduction. This in turn reduces food supplies for both local wildlife and people, which may have impacts for years to come. One example of

a localised event like this was the March 2023 oil spill in Poole Harbour, with a reported estimate of around 200 barrels worth of 'reservoir fluid' escaping an oil pipeline[3]. A report from December of 2023 details how, in the affected area, there were more than 25 different endangered[4], threatened, and protected species, highlighting how the populations of these species could have been negatively impacted in the case that this incident was more severe.

Another issue with the use of bitumen in roads is the process of cracking, which is used to isolate each individual fraction from crude oil. The processing of crude oil releases harmful gases[5], known as Volatile Organic Compounds (VOCs), into the atmosphere. VOCs can be harmful to human health, containing the carcinogenic gas Benzene, as well as the chemical Ethylbenzene. Ethylbenzene is classified as a potential carcinogen, which may increase the risk of the development of cancer, and can lead to severe and irreversible damage to the ears or kidneys. Some VOCs, such as methane, are potent greenhouse gases[6], contributing heavily to climate change: increasing global temperatures, altering weather patterns, and damaging ecosystems.

Living things are not the only things impacted by these increasing temperatures: so too is the infrastructure that makes up the world. A recent example would be the melting of the surface of the A143[7], a main road that connects parts of Norfolk and Suffolk together. The BBC reports that[7], in early July, the road surface melted near the village of Stanton, located in West Suffolk. This led to a road closure where the local council repaired the largest issues. Furthermore, the BBC stated that the same road surface had also been impacted in June after an earlier heatwave. These previous impacts suggest that the quality of the road surface may no longer be adequate, despite having been resurfaced in the September of 2025[7]. With rising temperatures in the summer months, similar incidents could become more common in the future. This would not only impact local people, who may be unable to travel in order to purchase supplies, but also the ability of Lorries to transport food and medical supplies, potentially impacting lives. In addition to these impacts, long term issues could occur. Some children would be unable to travel to school, which may cause a slight reduction in the education of local communities. This would likely lead to a lower regional average income, ultimately worsening quality of life in the area. The poor road conditions can also lead to damage inflicted on vehicles using the road, potentially costing thousands of pounds and even causing accidents, leading to a loss of life.

Through repairing these road surfaces, unfortunately, more asphalt is used, more bitumen must be harvested, and more oil is used. Each step in the process worsens the core problem that means that these repairs are required, creating a vicious cycle of emissions and consumption that destroys local climates one degree at a time. However, this situation is not entirely hopeless - there are things that could be done to slow the cycle. One of the things which could improve the health of these roads is the use of a different, lighter coloured material such as concrete. As this can be made in a different colour, the concrete will take longer to warm to as high a temperature as asphalt and will not melt or break down at regular temperatures. This is in contrast to the use of asphalt road surfaces which, due to their much darker colour, reflect less light. The light which is not reflected is instead absorbed by the road surface, increasing its temperature and causing it to melt. This may not be the best option, as concrete can be prone to crack after some time of use, causing road repairs to be required. Another potential option for road surfaces could be the use of gravel. Gravel is a naturally occurring resource, meaning that it would likely be much less expensive and require reduced processing compared to alternative materials. As it is both light in colour and a rock, gravel would

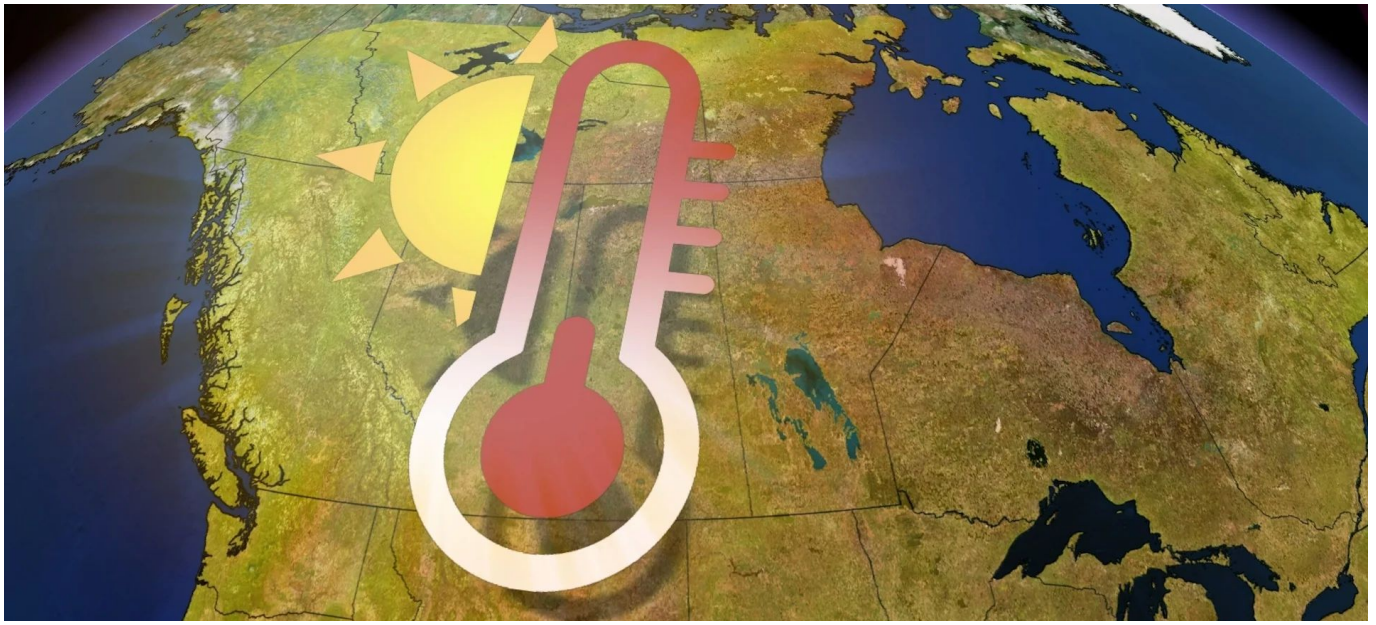
also have no risk of melting in the summer heat, as well as being suitable for Britain's cold and wet winters due to a decreased susceptibility to freeze-thaw weathering. Despite these benefits, gravel has its limitations. Roads with a gravel surface require more frequent maintenance than other roads with problems such as the formation of ruts, depressions in the road, which can prevent drivers from steering their vehicles. Furthermore, these solutions would necessitate the replacement of Britain's major road network, causing disruption to every person who may need to use them during their replacement and would cost the taxpayer a large amount of money. Another potential solution, which is already in place on some of the nation's busiest roads, is the use of a modern asphalt with polymer-modified binding agents. These agents increase the softening point of the material to 80 degrees Celsius as opposed to the usual 50 degrees reported by the RSTA[7], although they are not widely used due to their high price.

Roads in modern day Britain are, it could be argued, no longer able to keep up with the conditions that are experienced each summer due to the worsening effects of climate change. Whilst repairing them seems to return them to their usefulness for a time, this is only temporary. The more repairs that must be completed, the worse the climate becomes and the more ecosystems impacted. This cycle can be slowed but requires consideration from each perspective in order to have the best chance of success. Together, the people of the UK can quench the red-hot road to climate change and return to comfort.

By Z J Colton

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Climate Change: The Deadly Crisis Hiding In Plain Sight

When people hear the phrase ‘Climate change’, they often imagine disasters such as a devastating flood, searing forest fires or melting ice caps. Yet beneath these familiar headlines lies another crisis, one that receives far less attention: the widening impact of climate change on human health. From the increase of heat-related illnesses to the spread of mosquito-borne diseases in areas where they were once uncommon, healthcare systems are facing issues they were never designed to meet. Climate change can no longer be viewed only as an environmental issue; it is rapidly becoming one of the most defining health issues of the twenty-first century.

Heat-related mortality for people over 65 years of age increased by approximately 85% between 2000-2004 and 2017-2021 [1]. Behind this statistic are grandparents, loved ones, and neighbours whose bodies are struggling to keep up with these frequent and intense heatwaves. As temperatures rise, hospitals face an increasing number of vulnerable patients who require urgent care. This grave statistic reflects a harsher reality: the changing climate is no longer solely an environmental concern; it also places vulnerable populations at risk and creates immense pressure on public health systems. As the climate continues to change, environments are becoming more suitable for organisms that were once unable to thrive there. This can create conditions for certain disease vectors to spread beyond their original regions. For example, mosquitoes were found in Iceland for the first time after the country experienced record-breaking heat in 2025 [3]. Although the mosquitoes in Iceland were not found to be carrying diseases, it highlights a broader concern. As the climate continues to warm, disease-carrying mosquitoes could expand to regions that were unsuitable for their survival, which could increase the risk of more deadly cases such as malaria and dengue fever, a mosquito-borne viral disease. However, extreme heat is far from the only way climate change is reshaping human health.

As global temperatures continue to rise, air pollution follows suit. Higher temperatures intensify the factors that contribute to ozone formation, including heat, concentrations of precursor chemicals and methane emissions, further contributing to polluted air [2]. Consequently, as the air quality deteriorates, issues such as decreased lung function, respiratory problems, and cardiovascular diseases can increase. By 2050, researchers predict an additional 1,000 to 4,300 premature deaths annually due to the combined effects of ozone and particle pollution [2]. This illustrates the risk of serious health concerns, placing an ever-growing burden on healthcare

systems.

Climate change has placed large amounts of pressure on health services, creating challenges such as the rise in infectious diseases, the rising number of climate-related diseases and the disruption of healthcare infrastructure that stretch far beyond the efforts of health professionals. Protecting public health can no longer rely only on medical care; it now demands innovations in science, technology, engineering and mathematics to prevent health threats before they intensify.

AI has played an important role not only in outbreak detection but also in vaccine development and pandemic preparedness [4]. For example, an AI-based early warning system developed for India uses meteorological and health data to predict outbreaks such as dengue approximately two months in advance [5]. This is achieved by identifying the patterns between environmental conditions and prior outbreaks. Technologies such as this could help reduce the pressure on healthcare systems by giving health authorities more time to prepare for and respond to outbreaks, potentially protecting vulnerable populations from the health impacts of climate change. However, AI is not a perfect solution. Models trained using data from well-monitored regions may perform poorly in areas where surveillance is limited, potentially worsening existing health inequalities [5]. This illustrates that while technology can be useful, it cannot act as the sole solution for protecting communities from climate change. A climate-resilient healthcare system would need accurate data and collaboration across different areas of STEM.

A climate-resilient healthcare system is not only about treatments; it also involves the infrastructure that enables care to be provided. The World Health Organisation highlights the importance of designing healthcare facilities to withstand current and future climate risks [6]. The engineering sector is already contributing to this through technologies such as solar-powered photovoltaics, water pumps and telemedicine, which can strengthen the resilience of healthcare services [6]. During a climate emergency, normal electricity supplies may be disrupted, potentially affecting patient care. However, technologies such as solar photovoltaics can provide independent energy supplies, allowing healthcare facilities to continue functioning and providing essential care when it is needed most.

While engineering can help health services become more climate-resilient, it can also make them more sustainable. Healthcare supply chains account for an estimated 60-80% of a healthcare system's carbon footprint [6]. This highlights a serious contradiction: public health systems protect people from the health consequences of climate change, while their own activities can contribute to greenhouse-gas emissions. The engineering sector can help reduce this environmental impact through measures involving energy, buildings, water transport, the proper use and disposal of healthcare waste, and safer chemicals in medical products and devices [6]. Ultimately, creating health systems that are both resilient and sustainable can help ensure that essential care remains available to communities, including vulnerable and rural populations.

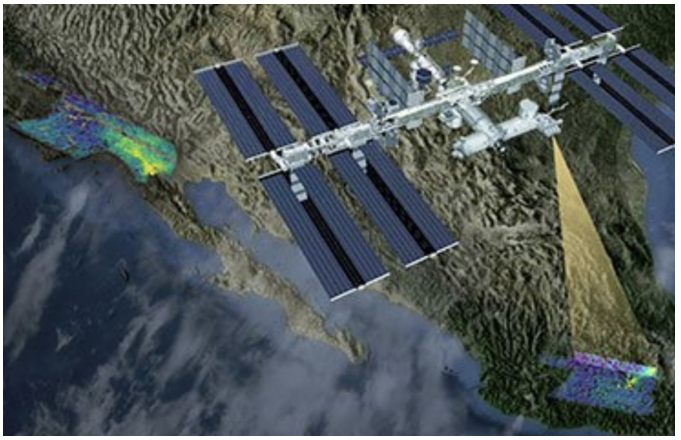
While technology can be a tool to help predict outbreaks, it is the science that helps explain why health risks are changing in the first place. For example, vector ecology is the study of how disease-carrying organisms, such as mosquitoes, interact with their environments and the pathogens they carry. It acts as an interface between environmental change and human health. Studying vector ecology helps researchers understand how environmental changes influence the spread of diseases such as malaria and dengue, as well as how their transmission can be controlled. As environmental conditions change, so too can the "distribution, abundance and

seasonal behaviour" of disease vectors such as mosquitoes and ticks [7]. This can create significant challenges for medical services as certain diseases might become more common in regions that were previously unsuitable for the vectors. Accordingly, scientists need to assess these changes and identify the areas which are becoming increasingly vulnerable to vector-borne diseases. Tools such as adaptive surveillance systems can be used by scientists to monitor the changing patterns in diseases, while innovative diagnostic tools can aid with the identification of infections [7]. Earlier identification of changing disease patterns and infections can allow health professionals to respond more quickly and make more informed decisions. This demonstrates that tackling the health consequences of climate change requires more than responding to outbreaks; it also requires scientific research that can help identify and prepare for emerging risks. Understanding the causes and geographical patterns of outbreaks can help scientists anticipate emerging risks and develop methods to protect communities.

Climate change is altering the face of human health, bringing growing threats to communities worldwide. However, STEM provides an opportunity to understand and respond to these changes. From predicting outbreaks through AI to improving healthcare infrastructure through engineering and using scientific research to understand changing disease patterns, these innovations can help healthcare move beyond simply reacting to crises and towards preparing for them. However, while these innovations have their uses, no single innovation can tackle a crisis of this scale. Through collaboration of STEM, healthcare, and communities, systems capable of withstanding the challenges ahead can be built. The greatest crisis may not be the one that can be seen, but the health crisis quietly developing alongside it. Climate change may be hiding in plain sight, but through STEM innovations and scientific understanding, the impact on human health can be identified and tackled.

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Model image depicting an OCO-3 satellite transmitting solar energy. Source: NASA Jet Propulsion Laboratory [1].

The Quest for a Greener World: Pioneering the Planet's Future Lands Through Space

It has been estimated that by 2100, $160 billion will need to be invested into preserving several yields of wheat, maize and soybeans due to persistent droughts that stem from global warming-intensified heat [2]. This encourages organisations to act urgently by engineering new plans in order to preserve and restore ravaged land, especially given the fact that approximately 45% of all habitable land is exploited for agriculture [3], proving that a large proportion of Earth will be affected if not managed.

Several associations have explored possibilities of utilising space technologies to mitigate impacts of anthropogenic climate change, whether it is through monitoring from outer space, or more ambitious projects such as space-based solar-power.

Through experiments and models, these organisations have proven that despite facing setbacks, space technology can majorly contribute to protecting land and making the ultimate quest for a greener world achievable.

Tracking Greenhouse Gases

Determined to make change, Dr Annmarie Eldering along with NASA adopted the pursuit of a new way to monitor climate change through space, particularly since most monitored information came from ground and ship-based measurements, meaning that a huge part of Earth was inaccessible to track [4], imposing the risk of not identifying major hazards in advance.

Together, Eldering and NASA's Jet Propulsion Lab innovated two Orbiting Carbon Observatory satellites: OCO-2 and OCO-3, to detect where carbon dioxide (CO_2) levels were strongest or the most apparent. It would use an instrument carrying spectrometers to measure the wavelengths of reflecting sunlight from a spot on Earth's surface [4]. The collected data would be compared to known patterns of absorption for CO_2 using precise measurements of light, indicating that detecting the gas' source was simpler. Therefore by comparing topographic maps of Earth with the CO_2 concentration map made, identifying outside factors that may have been contributing to the high CO_2 levels, such as volcanic eruptions from a mountainous area, was also easier.

However, since CO_2 contributes to environmentally-driven factors for climate change, detecting CO_2 alone would not identify areas affected by anthropogenic (human-induced) climate change.

In contrast, nitrogen dioxide (NO_2)is derived from human-induced causes of climate change, such as fuel combustion, that vastly augment the impact towards land. Therefore, discovering both CO_2 and NO_2 in an area would indicate that the root cause of the CO_2 is anthropogenic. Thus, regenerating polluted areas through implementing projects such as afforestation or urban greening would be attainable and efficient in minimising CO_2 concentrations. TROPOMI is a satellite that was brought into orbit by the European Space Agency together with

the Netherlands Space Office to measure NO_2 in the atmosphere [4]. It provides scientists with enough suitable data to combine both measurements of CO_2 from the OCO satellites and NO_2, allowing them to carry out actionable plans on earth such as introducing greener energy while receiving "direct feedback on how well they result in reducing CO_2 in the atmosphere" [4].

Obtaining Solar Power

While tracking air pollutants can positively contribute in planning other climate-alleviating initiatives, this method of improving the climate cannot generate clean energy to replace greenhouse gases. Alternatively, space-based solar power could be used to fuel future projects, using substantial and constant energy, as spacecraft would not be limited by the Sun's lack of direct sunlight during the night. Accordingly, modifying spacecraft to harvest solar energy would be significantly more effective in terms of mollifying climate-induced impacts, by allowing renewable energy to be transmitted directly to degraded regions through grids connected to power lines.

This is what the OHISAMA team planned to do; launch a satellite that can convert light from the sun into microwaves received by antennas or grids on Earth through wireless transmission [5]. Afterwards, hardware would convert the microwaves into electricity to be dispersed and utilised.

Scientists under the OHISAMA and Japan Space System teams initially tested this project by launching a satellite, containing transmission panels, 7km above sea level [5]. Subsequently, a 64-meter parabolic antenna was set up at the Usada facility, known for tracking spacecraft, in an effort to receive 720 watts of energy transmitted wirelessly [5]. The transfer was deemed successful, reassuring scientists that this project would be feasible and likely to succeed at even greater altitudes of up to 450km.

Under the Caltech institute, a university recognised for their innovative scientific work with NASA'S Jet Propulsion Lab, an experimental aircraft was then launched into space in January 2023 that tested the reliability of using solar power through wireless transmission whilst being several miles away.

To initially build such an ambitious probe would have involved careful consideration and planning as to avoid failure. As stated in [6], the transmitters used to beam energy back down onto Earth would need to meet a certain criteria: being flexible for adequate portability, lightweight to minimise fuel usage and inexpensive.

One way these targets were met was by modifying the interfaces among transmitters to control the alignment of converted microwaves, allowing for meticulous precision when relaying the microwaves [6]. Additionally, the model holds receiver arrays for receiving and converting energy into direct current electricity, meaning that less energy is wasted and therefore is valuable when considering the financial constraints kept throughout the experiment.

Ultimately, receivers on a roof were able to successfully detect transmitted energy from the aircraft, demonstrating how productive solar power can truly be. This achievement encouraged scientists and engineers to invest into developing a ground for solar power, such as the Space Solar Power Project (SSPP) reported to soon 'deploy a constellation of modular spacecraft' [6] carrying out the same job of converting solar energy into microwaves as the OHISAMA team's satellites.

Furthermore, through the installation of SSP replacing finite energy sources, degraded land could be directly protected; converted solar energy would be accurately transmitted back onto a grid located near areas in need of aid.

Consequently, clean energy would power machinery to restore land, akin to how energy from solar panels can straightforwardly power a house.

Such an advantage would limit the amount of polluting energy sources being utilised, whilst simultaneously protecting lands and permitting environmental development. Hence, it would reduce greenhouse gas emissions to ultimately control impacts of climate change.

Adapting to Budgets

Despite all these positive indicators for advancements in space-based solar-power, NASA claims that the expenses put into generating energy, building and launching systems were greater than other sustainable methods; using spacecraft to generate 1 kilowatt per hour would cost 61¢ compared to only 2¢ using hydropower [7]. This portrays space solar power (SSP) as one of the most expensive methods of preserving and developing the planet when prolonged for a long time, creating a major setback when balancing budgets; according to the third table in [8], SSP that would come from systems with enough research and development completed would sell for 9.4¢ per kilowatt-hour, a price that over time would lead to a major loss in profit.

However, following the possibility of the Space Solar Power Project using satellites, several adjustments such as miniaturisation could be made to control expenditures whilst potentially upgrading the effectiveness of satellites.

NASA's CubeSat initiative follows the mission of launching downsized satellites that can undertake several investigations as well as demonstrations for teaching in space [9]. CubeSats have been used as the blueprint for multiple successful projects, such as the CubeSat-compatible Compact Thermal Imager which encompasses modern detectors capable of reacting to infrared radiation (infrared detectors). Through these detectors, the CubeSats have not only captured more than 15 million infrared images on Earth, but also detected wildfire activity and monitored crop conditions [10], clearly demonstrating the immense capability such small satellites carry.

For a project as demanding as space-based solar power, the same rewarding miniaturisation concept could be applied in order to drastically reduce expenses while still maintaining function and reliability.

Additionally, the largely renewable supply of energy offered through SSP is sustainable enough to outweigh other power sources, as it is not dependent on intermittent environmental factors such as water for hydropower or biofuel derived from dead organic matter called biomass. Here, space-based solar power becomes more relevant when considering how hydropower and biomass may not be sufficient in the future; recurring droughts resulting from increased temperatures, mentioned in [2], are increasingly depriving land of water. Hence, there would be insufficient water to encourage hydropower generation, concurrently limiting organic materials to grow and produce biomass energy.

Ultimately, it demonstrates that investing a considerable amount of money into space-based solar power to ensure efficient impacts is worthwhile compared to cheaper yet infrequent sources such as tidal energy, especially when managing a challenging matter such as climate change.

By Alexandra Ivli

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Reinventing the Oil Rig: Wind, Water and Carbon for Climate Solutions

Oil rigs are the major symbol of the fossil fuels era, but imagine if they were used as part of the green technology to move beyond the use of these environmentally harmful materials… Offshore oil rigs are platforms situated in open waters used to extract a widely utilized type of hydrocarbon: petroleum [1].

Petroleum is a complex liquid mixture composed mainly of carbon and hydrogen atoms of fossil origin, meaning it is found inside sedimentary rocks of the earth's surface. This compound has been essential for the development of modern society from the Industrial Revolution to the present day because it represents the main energy resource and a key raw material for modern industry [2].

After petroleum is extracted, a combustion reaction takes place in which a fuel reacts with oxygen to release energy in the form of heat and light. The downsides of this process are that the reaction releases fine toxic particles into the air, which are harmful to human health, as well as CO_2, a greenhouse gas that traps heat in the atmosphere and intensifies the greenhouse effect. Both of those pollutants heavily contribute to climate change and global warming [3].

A more sustainable alternative would be to replace fossil fuels with electrochemical processes. These processes involve reactions that convert energy between chemical and electrical forms, as in electrolysis. A prime example of how this could be achieved is through wind power 4, a great source of renewable energy for a structure in open waters subject to strong winds. Wind power would be generated by wind turbines and the electricity produced would be used to power electrolysis reactions, processes in which electricity is used to drive chemical reactions. In addition to being windswept, oil rigs have an almost inexhaustible water supply. The electrolysis reaction that would take place is: $2H2O \rightarrow 2H2 + O2$

The reason why electricity generated from wind power is needed for this apparently simple reaction is that H2O is a very stable and strong molecule; this means that a large quantity of energy is needed to break its bonds. Among the products of this reaction, H2, molecular hydrogen, would be obtained. H2 represents the lightest and most abundant element in the entire universe [5] and it is also known as an energy carrier as it stores energy that is released through a combustion process. Unlike fossil fuels, this process releases only water vapor and clean energy, with no CO2 emissions.

Even if this sounds like the ultimate solution to climate and environment problems, industries still need carbon to produce plastics, medicines, materials, fertilisers and so on. Choosing to avoid petroleum, roughly 85% carbon, CO2 could be used instead. It is the same compound driving the decline of the planet's health that is being targeted for elimination. However, because it is such a stable compound, it can be classified as a valuable raw material and not simply a pollutant. Due to the presence of carbon in the CO2 molecule, it can be converted into useful carbon-based products through reduction reactions. These reactions can be carried out electrochemically or catalytically. In an electrochemical reduction, electricity is used to drive the reduction of CO2, whereas in a catalytic reduction, a catalyst facilitates the reaction by lowering its activation

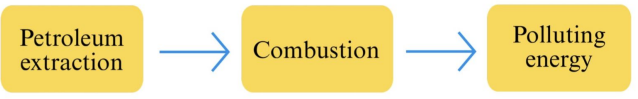
energy making it easier for the reaction to occur.

The process of this electrochemical reaction uses electric energy to transform CO_2 into fuels and higher-value chemical products. If the electric energy used is obtained from renewable sources, this process allows the recycling of carbon, reducing greenhouse gas emissions and generating high-value molecules. The most common high-value molecules to obtain are carbon monoxide, methanol, ethylene and formic acid [6]. All of them contain carbon atoms, implying they can be classified as carbon products:

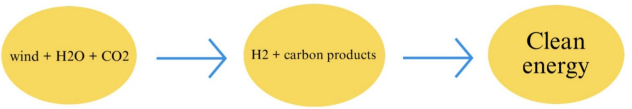
Molecule name	**Formula**
Carbon monoxide	**C**O
Methanol	**C**H_3OH
Ethylene	$\mathbf{C_2}H_4$
Formic acid	H**C**OOH

Carbon itself is not the real problem for the world's climate difficulties but it could be used to generate carbon products without continuing to extract petroleum from the seabeds. To get the product, catalysts are needed. They work as tools that speed up the reaction without being consumed in the process and can be implemented in the chemical reaction to also direct it to the desired product.

Adopting these changes could transform an offshore oil rig process from:



to

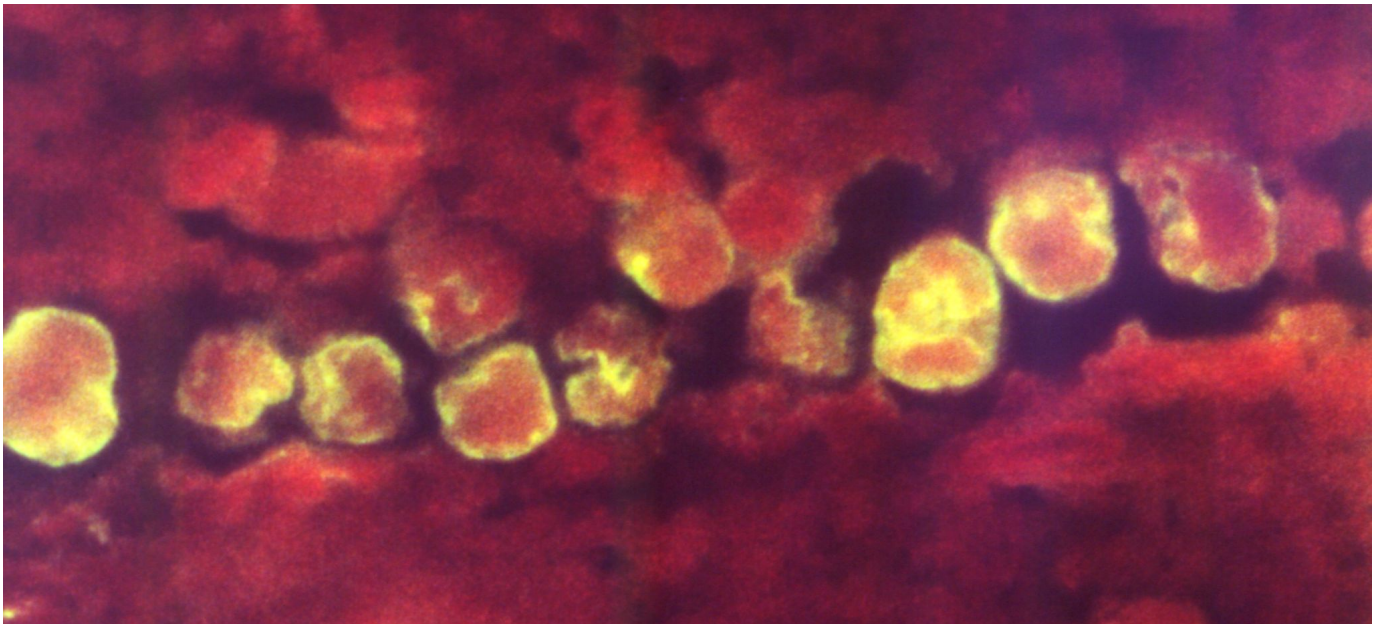


A hypothetical difficulty could present itself in the form of the renewable source of energy: the wind. Wind power is classified as an intermittent power supply, as well as solar energy, meaning there could be periods where it will not be possible to obtain the quantity of energy needed. Other renewable sources, such as hydropower, can provide a more stable energy supply, although its availability depends on geographical and environmental conditions. When renewable sources cannot guarantee a continuous energy supply, a possible candidate could be nuclear energy. Nuclear energy is a rapidly developing form of energy that is continuously being tested and improved. Its strengths include being a dispatchable form of energy, producing a relatively low quantity of waste, and generating energy with almost no carbon emissions as its production process does not release fumes in the air. These characteristics make nuclear energy a potential candidate as a substitute for wind power [7].

Whether the answer lies in wind, nuclear or something in between, the real challenge is not simply to substitute petroleum with a new source of energy but to carefully consider the impact that transforming such a critical energy-producing system would have on climate and environment in order to pursue a more sustainable future.

By Asia Pansera

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Climate Change and the Expanding Distribution of *Naegleria fowleri*: The Brain-Eating Amoeba

In October 2022, a young boy under the age of 18 died after contracting a rare infection which is 97% fatal in the US [1]. Medical officials believe he may have encountered the organism at the Kingman Wash area on the Arizona side of Lake Mead, near Hoover Dam [2]. The cause was *Naegleria fowleri*, a microscopic amoeba historically confined in warm freshwater in the southern region of the United States [3]. This was the first fatality linked to the amoeba at Lake Mead and third case of that year in the United States [2]. While this incident appears isolated, it aligns with an observed trend. Rising global temperatures could be associated with *N. fowleri's* expanding geographical range as more areas with favourable environmental conditions appear. As cases of *N. fowleri* appear in new areas, the lack of effective treatment becomes an increasing problem.

Biology and Infection

Naegleria fowleri is an amoeba found in soil and warm freshwater lakes, rivers, ponds and hot springs around the world. However, its distribution is concentrated around specific regions [4]. The life cycle of *N. fowleri* consists of three stages: the trophozoite, flagellate and cyst forms. The trophozoite is the active stage responsible for infection, while the cyst allows the organism to survive when environmental conditions become less favourable. *N. fowleri* is thermophilic, meaning it grows best in warmer conditions. If water containing the amoeba enters through the nasal passage to the brain, it can cause an infection called primary amebic meningoencephalitis (PAM). PAM cannot be contracted by ingestion and is non-communicable. Fewer than 10 people acquire PAM a year in the United States — the majority of them die from it. Between 1962 and 2024, there were 167 reported cases of PAM in the United States. Only four have survived [5]. The pathogenicity of *N. fowleri*, which refers to its ability to cause disease, can result from nerve damage in the CNS due to cytolytic chemicals or through brain consumption. Additionally, *N. fowleri* is believed to inhibit the immune system as the infection progresses which contributes to the high fatality rate [4]. The symptoms of the infection often resemble viral or bacterial meningitis, making misdiagnosis a recurring issue. The question of whether warmer conditions allow *N. fowleri* to expand into new regions still remains.

Thermal Drivers of Growth and Distribution

For a thermophilic amoeba, temperature is one of the most significant environmental factors influencing *N. fowleri's* growth and distribution. *N. fowleri* is able to grow

between 25°C and 42°C, while persisting for short durations at water temperatures above 45°C, usually for 24-96 hours. *N. fowleri* does not replicate at 20°C [6, 7]. As temperatures increased towards 40°C, greater activity and shorter generation times were observed. While in cyst form the amoeba can survive in cold temperatures below 10°C [6], in this form the amoeba is unable to attach to and invade a host. Rising atmospheric air temperatures are accompanied by increases in freshwater temperature, although the magnitude of warming is also affected by groundwater inputs, riverbed characteristics, streamflow and discharge [8]. Extreme heat events — like droughts and heatwaves, and lower water levels can further increase freshwater temperatures, potentially extending periods where conditions are suitable for thermophilic organisms [9, 10]. Together these factors demonstrate how climate-driven changes may influence the geographical and seasonal availability of environments suitable for *N. fowleri* growth.

Evidence of Geographic Expansion

Although warmer conditions may increase environmental suitability, evidence is required to determine whether this translates into geographical expansion. Evidence of PAM cases can help determine whether infections are appearing outside of historically recognised regions. To investigate geographical distribution of PAM cases in the United States, Gharpure et al. [3] have analysed the trends in locations of PAM incidents. Evidence from the study suggests PAM is not restricted to traditionally recognised areas in the United States, with reported cases across a wide geographical range between 1978 and 2018, even though *N. fowleri* has been historically reported in predominantly southern regions of the United States. This suggests that suitable conditions exist outside of the previously acknowledged areas, potentially influenced by the changing climate. However, the geographical distribution of PAM cases does not indicate climate change as the sole driver of expansion [3]. In other words, human infection data can only show where people got sick, not where the amoeba actually lives. A 2026 study by Shikany et al. [11] investigated the presence of *N. fowleri* in thermally impacted recreational waters including five national parks. Environmental sampling was combined with laboratory identification, genetic confirmation and statistical analysis to determine the presence of *N. fowleri*; the amoeba was detected in 34% of samples. It was found specifically in three out of five of the national parks [11]. The study does not indicate the amoeba was new to the areas or the influence of climate change, only current presence. Gharpure et al.'s [3] study was restricted due to the rarity of infection leading to an indirect inference of the amoeba's spread. The 2026 study by Shikany et al. [11] helps address the limitation, providing us with the locations where the organism was detected through direct water testing, not just statistical implications from past cases.

Diagnostic and Treatment Challenges

Despite its rarity, PAM is exceptionally difficult to treat [12]. The geographical expansion may become a genuine health concern in new areas where the public have limited awareness and clinicians lack diagnostic familiarity. Symptoms of PAM appear between 1 and 12 days post-exposure with the severity of the symptoms rapidly progressing to seizures, coma and death [12]. The quick progression often leaves clinicians little time to correctly identify and treat the infection. Additionally, early symptoms of PAM are usually non-specific: fever, headache, nausea, and fatigue. As a result, many patients are often misdiagnosed and treated for the incorrect infection [12]. While there are no consistently effective treatments for PAM, Centers for Disease Control and Prevention (CDC)-recommended therapy includes a combination of drugs such as amphotericin B, miltefosine, fluconazole, rifampin, azithromycin, and dexamethasone. However, these treatments are accompanied by substantial limitations, including high toxicity and poor blood-brain

barrier penetration which impedes effective CNS delivery [13]. Due to the rarity of PAM and high mortality rate, there is a lack of clinical trials [13], resulting in treatment guidance from the CDC being based on survivor case reports or in vitro studies [12]. Televised precautions during times of year when temperatures are optimal for *N. fowleri*, especially for recreational water users, could reduce PAM infections.

The 2022 case in Lake Mead is but one concrete example of how climate change is reshaping disease geography, driven by broader ecological disruptions such as prolonged droughts and rising water temperatures. While current evidence suggests wider distribution, though climate change cannot yet be considered the sole reason, this ecological shift underlines the urgent need for public health safeguards into freshwater management. As warming trends progresses, infections in regions with no prior recorded cases become an issue deserving of far more attention — not just from researchers tracking the amoeba's spread, but from clinicians who need to consider PAM in areas where it is no longer absent.

By Victor Adetula

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Thank You!

This concludes the competition issue of the Penrose Magazine. Thank you so much to everyone who participated in this competition and to all of the Penrose Team who dedicated time to editing, marketing, and formatting.

Special thank you to our judges who dedicated time to help us decide the competition winners:

Christopher Lowe	Emeritus Professor of Biotechnology at Cambridge, OBE, FREng
Ettore Biondi	Assistant Professor of Geophysics, Stanford University
Alessandro Alabastri	Assistant Professor of Electrical and Computer Engineering Director, Applied Physics Graduate Program Core Member, Smalley-Curl Institute, Rice University
Mahan Ghafari	Wellcome Trust Early Career Research Fellow, Ghafari Research Group; Pandemic Sciences Institute; Lincoln College, Department of Biology, Oxford University
Brenda Boonabaana	Assistant Professor in the Department of Geography and the Environment, University of Texas at Austin

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